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A fuzzy-stochastic optimization model for the intermodal fleet management problem of an international transportation company

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ABSTRACT

In this paper, a fuzzy-stochastic optimization model is developed for an intermodal fleet management system of a large international transportation company. The proposed model integrates various strategic, tactical and operational level decisions simultaneously. Since real-life fleet planning problems may involve different types of uncertainty jointly such as randomness and fuzziness, a hybrid chance-constrained programming and fuzzy interactive resolution-based approach is employed. Therefore, stochastic import/export freight demand and fuzzy transit times, truck/trailer availabilities, the transport capacity of Ro-Ro vessels, bounds on block train services, etc. can also be taken into account concurrently. In addition to minimize overall transportation costs, optimization of total transit times and CO_2 emission values are also incorporated in order to provide sustainable fleet plans by maximizing customer satisfaction and environmental considerations. Computational results show that effective and efficient fleet plans can be produced by making use of the proposed optimization model.

KEYWORDS

Fleet planning; fleet sizing and composition; intermodal transportation; empty vehicle repositioning; fuzzy-stochastic programming; case study

1. Introduction

Fleet planning problems need to be solved by handling various complex decisions simultaneously such as transportation mode selection, load planning, outsourcing, frequency of maritime and railway cruises, empty vehicle repositioning, fleet sizing/composition, fleet allocation and expansion/reduction decisions (Bierwirth, Kirschstein, and Meisel 2012; Meisel, Kirschstein, and Bierwirth 2013). However, most of the available studies in the literature are concentrated on single mode fleet management from the perspectives of logistics companies or production firms that generally utilize road freight transportation. There are a few papers on intermodal fleet sizing/composition problems when compared to the unimodal transportation systems (cf. Baykasoğlu et al. 2019). Moreover, fleet planning problems are much more complicated in intermodal transportation systems than unimodal road freight transport which considers only the trucking part of the overall transportation system. As highlighted by Baykasoğlu and Subulan (2016), load planning or intermodal freight transportation planning is one of the most vital problems since it

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influences the other planning decisions directly – the outputs from the load planning problem are used as inputs by all of the other fleet planning issues. Unfortunately, most of the available studies in the literature generally handle these problems in a separate way. Additionally, as mentioned by SteadieSeifi et al. (2014), no paper deals with the dynamic positioning and repositioning problems of multiple assets or resources such as trucks, trailers, containers, etc. In reality, the empty resource repositioning issues should also be included as part of fleet planning problems. Although the repositioning of these empty resources does not contribute to transportation profits directly, it is able to increase the customer service level by decreasing waiting time. SteadieSeifi et al. (2014) also reported that simultaneous planning of multiple resources, i.e. trucks, trailers, containers, drivers, etc. and integration of backward flows (empty truck repositions) into forward flows (loaded truck movements) will provide significant time saving and cost reduction. As also highlighted by Caris, Macharis, and Janssens (2008) there is little in the research literature which covers various time horizons and integrates the decisions of more than one decision maker. Based on these motivations, this paper presents a holistic fleet planning model by incorporating the perspectives and objectives of several decision makers with different levels of decision. The proposed intermodal fleet planning model incorporates the following strategic, tactical and operational planning decisions:

Strategic fleet planning decisions:

- Truck and trailer fleet size optimization,
- Truck fleet composition (type of truck plates – European and Turkish plates),
- Truck and trailer fleet expansion (purchasing new trucks/trailers) and reduction decisions (sales of second-hand trucks/trailers).

Tactical fleet planning decisions: Intermodal freight transport planning (multi-mode load plan),

- Loaded truck movements, truck and trailer allocations to different road transport routes,
- Transportation mode and service type selection,
- Outsourcing,
- Periodic truck and trailer inventories on the intermodal logistics network.

Operational fleet planning decisions:

- Internal empty truck and trailer repositioning,
- External empty truck and trailer repositioning,
- Load consolidation at transshipment seaports and railway stations.

Coping with uncertainties such as stochasticity, dynamism and fuzziness related to the elements of an intermodal transportation system (stochastic freight demands, fuzzy transit times, mode capacities, etc.) can be seen as the other research challenges. For this reason, it is proposed in this paper that monthly import/export freight demands of the order countries are not exactly known and considered as stochastic parameters since they have randomness in their nature. Furthermore, all of the cost parameters, transport

capacities by marine and rail transports, transit times, number of working days at each month, truck and trailer availabilities, CO_2 emission values of different transport modes etc. are treated as fuzzy parameters and described by possibility distributions because of their epistemic uncertainty resulting from unavailability or incompleteness of the historical and insufficient data etc.

In light of this information, the proposed holistic fleet planning model is formulated as a fuzzy-stochastic mixed-integer programme for the examined multi-objective, multi-mode, multi-stage, multi-resource, multi-period fleet planning problem. Apart from the existing literature, both of the import/export transportation operations are considered in the load planning model component. In addition to optimize the size of the truck fleet with different plates, trailer fleet size is also determined. The repositioning decisions on both empty trucks and trailers are made by considering the periodic truck/trailer inventories. In addition to minimize overall transport costs, the proposed model also aims to minimize total transit time and environmental impact resulting from CO_2 emissions by different modes of transport. It is targeted to achieve maximal customer satisfaction or service level by minimizing the total transit times. A weighted additive fuzzy goal programming approach is applied to handle these conflicting objectives simultaneously and obtain compromise solutions under different uncertainty levels. Finally, a real-life application is presented for a large international transportation company in Turkey for testing validity and practicality of the proposed model. Since a comprehensive literature review on single and multi-mode fleet planning problems has recently been presented by Baykasoğlu et al. (2019), this paper does not include a literature survey. To the best of our knowledge and the previously published literature review paper, a similar fleet planning model which is studied through this paper is not encountered in the current literature.

The rest of this paper is structured as follows: Section 2 presents the details of the intermodal fleet planning problem with its mathematical model formulation. A hybrid chance-constrained stochastic programming and fuzzy mathematical programming-based solution approach are described in Section 3. In Section 4, computational results of the real-life application for the multi-objective fuzzy-stochastic fleet planning model are carried out and presented. Finally, concluding remarks and future work are presented in Section 5.

2. Intermodal fleet planning problem and its mathematical model formulation

There are three different transport services in the examined intermodal fleet planning problem. Based on the given fixed logistics configuration, loaded truck/trailer positions and empty truck/trailer repositions between the different locations on the intermodal transportation network can be demonstrated by Figures 1–3. According to Figure 1(a), loaded truck positions and internal empty truck repositions can be performed between the seaports in Europe and order countries. In addition, external empty truck repositions can be realized among the seaports themselves in Europe and between order countries, railway stations and seaports in Europe. Moreover, there are also loaded truck positions and internal empty truck repositions between the railway stations in Europe and order countries as shown in Figure 1(b). Additionally, external empty trucks can be transferred from the other railway stations, seaports in Europe and order countries in order to meet the truck requirement of a railway terminal. Furthermore, loaded and empty truck flows

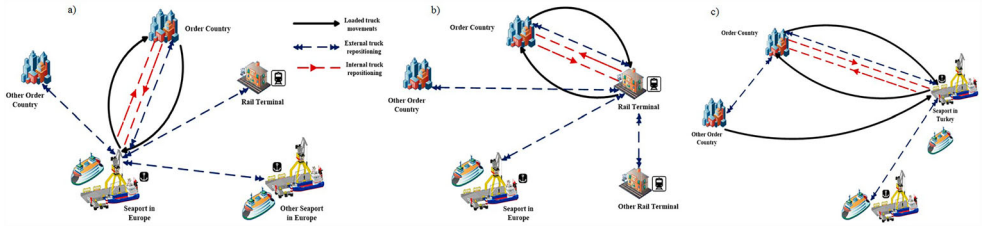


Figure 1. Loaded/empty truck positions/repositions: (a) from/to seaports in Europe, (b) from/to railway stations in Europe, (c) from/to seaports in Turkey.

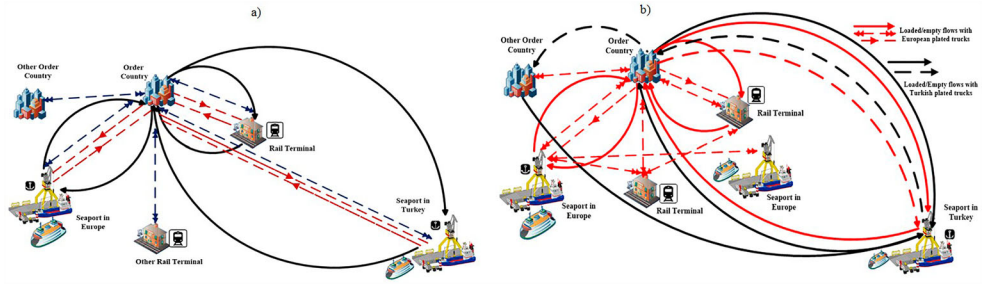


Figure 2. Loaded/empty truck positions/repositions: (a) from/to order countries in Europe, (b) by European and Turkish plated trucks on the intermodal logistics network.

from/to a seaport in Turkey can be visualized as in Figure 1(c). The loaded and internal empty flows can be performed by both European and Turkish plated trucks between seaports in Turkey and order countries. External Turkish plated empty trucks can also be repositioned among the seaports themselves in Turkey. European/Turkish plated trucks

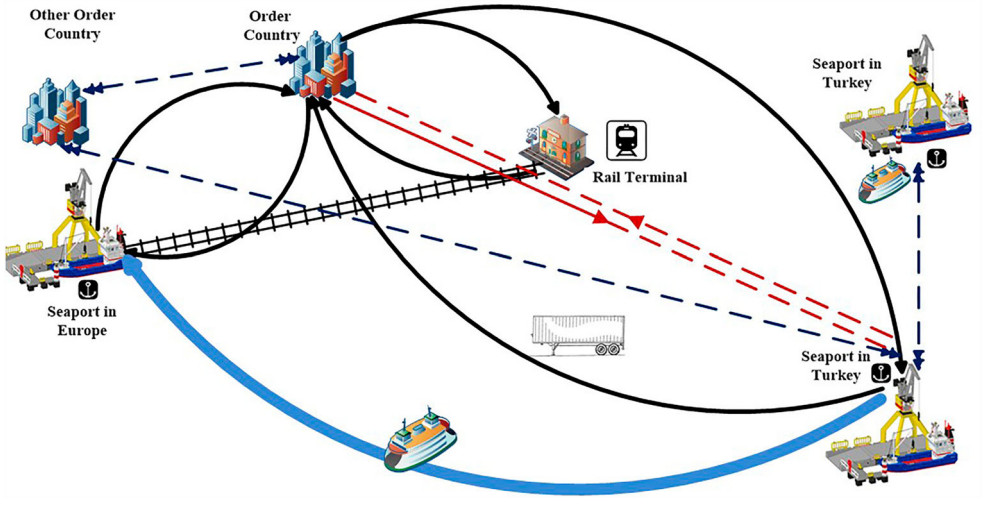


Figure 3. Loaded/empty positions/repositions of trailers between seaports in Turkey and order countries in Europe.

can be externally redistributed from order countries to seaports in Turkey. Turkish plated trucks are also externally repositioned with their empty trailers among the order countries.

Figure 2(a) presents the loaded and empty truck flows from/to the order countries in Europe. In detail, the loaded truck positions and internal empty truck repositions are realized between the order countries and railway stations, seaports in Europe and Turkey. Besides, European plated empty trucks can be externally repositioned among the order countries. The entire loaded and empty flows by European and Turkish plated trucks are summarized as in Figure 2(b). It is clearly seen from this figure that both the loaded and empty flows of Turkish plated trucks are relatively limited when compared to European plated trucks. Finally, loaded trailer positions and empty trailer repositions between the seaports in Turkey and order countries in Europe are depicted in Figure 3. Although the loaded trailers can be shipped by road, marine and rail transports, empty flows of trailers can be realized by only road transport. The empty trailers can also be externally repositioned among the order countries, seaports themselves and between the seaports in Turkey and order countries.

In the modelling stage of the above intermodal fleet planning problem, the deterministic mixed-integer programming (MIP) model is initially developed with four fleet planning model components. In detail, multi-mode load planning or intermodal freight transportation planning model is first proposed afterwards extended by incorporating the truck/trailer fleet size and composition, loaded and empty truck/trailer movements, truck/trailer inventories on the logistics network and fleet expansion and reduction decisions. In detail, each model component may contain some outputs of the former components and generate the inputs of the latter components. However, it should be noted here that these components do not constitute an iterative process. The first model component, i.e. multi-mode load planning of the proposed intermodal fleet planning framework is extensively discussed by Baykasoğlu and Subulan (2016). For this reason, subsequent components are the main focus of this paper instead of the first model component. In the second model component, truck/trailer fleet size optimization, fleet composition and fleet allocation are carried out. In contrast to the existing studies in the literature, both the truck and trailer fleet size optimization is intended in this paper. Fleet size of vehicles is determined for each time period by considering the loaded truck/trailer positions, empty repositions, transit times and length of the period and truck/trailer availabilities. In detail, quantities of loaded truck positions are obtained from the solution of the first component. In addition to these loaded truck movements, road-marine-rail transit times between the origin and destination pairs, the monthly duration (in hours) and the percent of time for truck/trailer availabilities are needed so as to determine the appropriate fleet size.

Composition of the truck fleet is also determined by considering different types of identification plates such as European and Turkish plated vehicles. Actually, because of the cabotage laws in Europe, this is an important issue for realizing the loaded and empty truck movements with different types of plates. An appropriate fleet sizing/composition and fleet allocation can also provide minimal empty truck/trailer repositioning (Sayarshad and Ghoseiri, 2009; Sayarshad and Tavakkoli-Moghaddam, 2010). For a better understanding of fleet sizing, fleet allocation and internal empty truck repositioning concepts, an illustrative numerical example is presented in Figure 4. If the trucks can be

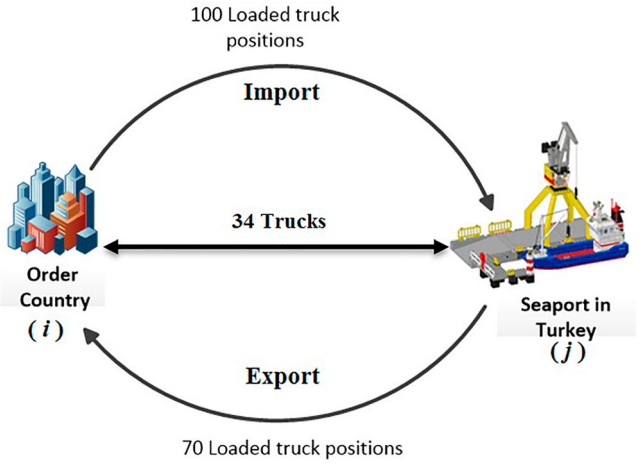


Figure 4. An illustrative example for fleet sizing and allocation by considering empty repositions.

used more than once in a specific time period, parameters related to travel times, length of the planning period and availability of the sources will be essential as well as the loaded truck positions in fleet sizing. Suppose that the number of loaded truck movements between nodes i and j are 100 and 70 positions. The travel time between these nodes is 100 time-units per position. The planning horizon is set to one month and there are 25 working days available in a month. In fact, each truck can perform six trips among these nodes throughout the planning period. Moreover, it is assumed that trucks are always ready for transportation operations or available 100% at a given time period. In this case, the minimum number of required trucks at arcs (i, j) and (j, i) can be calculated by the following equations (Baykasoğlu et al. 2015):

$$\text{Truck}_{ij} + \text{Truck}_{ji} \geq 2 * \left[\frac{\max \{\text{Time}_{ij} * \text{Position}_{ij}, \text{Time}_{ji} * \text{Position}_{ji}\}}{\text{Period Length} * \text{Truck Availability}} \right] = \frac{\text{Time}_{ij} * \text{Position}_{ij} + \text{Time}_{ji} * \text{Position}_{ji} + |\text{Time}_{ij} * \text{Position}_{ij} - \text{Time}_{ji} * \text{Position}_{ji}|}{\text{Period Length} * \text{Truck Availability}} \tag{1}$$

$$\text{Period Length} = \text{Number of working days} * \text{duration (in hours)} \tag{2}$$

According to the above formulations, the total number of required truck fleet size at nodes i and j can be computed as follows:

$$\text{Truck}_{ij} + \text{Truck}_{ji} \geq \frac{(70 * 100 + 100 * 100 + |70 * 100 - 100 * 100|)}{600 * 1.0} \approx 34 \tag{3}$$

After determining the appropriate truck fleet size, the allocation of these trucks between the nodes i and j is so vital in terms of solution quality. In detail, amounts of empty truck repositions are directly related to initial positions of these trucks. In other words, the total number of 34 trucks should be allocated accurately to the routes among these nodes or arcs (i, j) and (j, i) so as to transport the planned import/export freights

within the given time period. Moreover, fleet allocation is essential due to its significant effects on total transportation costs because a large number of empty truck repositions may arise when the truck fleet is not allocated effectively (Sayarshad and Tavakkoli-Moghaddam, 2010).

The required number of empty truck repositions between nodes i and j can be determined as follows:

$$\text{Reposition}_{ji} \geq \text{Position}_{ij} - \text{Position}_{ji} - \text{Truck}_{ij} \tag{4}$$

$$\text{Reposition}_{ij} \geq \text{Position}_{ji} - \text{Position}_{ij} - \text{Truck}_{ji} \tag{5}$$

Finally, after fleet allocation and empty truck repositions, periodic truck inventories at the end of the planning period can be formulated as follows:

$$\begin{cases} \text{Inventory}_i = \text{Truck}_{ij} - \text{Position}_{ij} + \text{Position}_{ji} + \text{Reposition}_{ji} - \text{Reposition}_{ij} \\ \text{Inventory}_j = \text{Truck}_{ji} - \text{Position}_{ji} + \text{Position}_{ij} + \text{Reposition}_{ij} - \text{Reposition}_{ji} \end{cases} \tag{6}$$

Furthermore, the physical representation of fleet allocation and impact of allocation decisions on solution quality can be clarified by the following three cases, i.e. perfect, poor and comparatively better allocations:

Case-1: A perfect fleet allocation without any empty truck reposition: 30 trucks are available on arc (i, j) and four trucks on arc (j, i) at the beginning of the planning horizon.

$$\text{Reposition}_{ji} \geq 100 - 70 - 30 = 0 \qquad \text{Inventory}_i = 30 - 100 + 70 + 0 - 0 = 0,$$

$$\text{Reposition}_{ij} \geq 70 - 100 - 4 = -34 \qquad \text{Inventory}_j = 4 + 100 - 70 + 0 - 0 = 34$$

Truck inventory at Order Country (i)	**Trip-0** Number of import positions	Truck inventory at Seaport in Turkey (j)
30 →	100	
	Number of export positions	
	70 ←	4

Truck inventory at Order Country (i)	**Trip-1** Number of import positions	Truck inventory at Seaport in Turkey (j)
4 →	70	
	Number of export positions	
	66 ←	30

Truck inventory at Order Country (i)	**Trip-2** Number of import positions	Truck inventory at Seaport in Turkey (j)
30 →	66	
	Number of export positions	
	36 ←	4

Truck inventory at Order Country (i)	**Trip-3** Number of import positions	Truck inventory at Seaport in Turkey (j)
4 →	36	0
	Number of export positions	
0	32 ←	30

Truck inventory at Order Country (i)	**Trip-4** Number of import positions	Truck inventory at Seaport in Turkey (j)
30 →	32	0
	Number of export positions	
0	2 ←	2 + 2

Truck inventory at Order Country (i)	**Trip-5** Number of import positions	Truck inventory at Seaport in Turkey (j)
2 →	2	0
	Number of export positions	
0	0	30 + 2

Truck inventory at Order Country (i)	**Trip-6** Number of import positions	Truck inventory at Seaport in Turkey (j)
0	0	0
	Number of export positions	
0	0	32 + 2 =34

In this case, it is clear that there is no need to reposition any empty trucks among the nodes. The number of empty truck repositions is explicitly associated with initial truck positions.

Case-2: A poor fleet allocation which causes maximal empty truck repositions: zero trucks are available on arc (*i, j*) and 34 trucks on arc (*j, i*) at the beginning of the planning horizon.

Truck inventory at Order Country (*i*)	**Trip-0** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
0	100	
Number of export positions		
	70 ←	34

Truck inventory at Order Country (*i*)	**Trip-1** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
34 →	100	
Number of export positions		
	36	0

Truck inventory at Order Country (*i*)	**Trip-2** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
0	66	
Number of export positions		
	36 ←	34

Truck inventory at Order Country (*i*)	**Trip-3** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
34 →	66	0
Number of export positions		
0	2	0

Truck inventory at Order Country (*i*)	**Trip-4** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
0	32	0
Number of export positions		
0	2 ←	34

Truck inventory at Order Country (*i*)	**Trip-5** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
Empty 30 + 2 →	32	0
Number of export positions		
0	0	2 **Awaiting**

Truck inventory at Order Country (*i*)	**Trip-6** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
0	0	0
Number of export positions		
0	0	32 + 2 =34

In the second case, in order to conduct loaded truck positions during the fifth trip, an additional 30 empty truck repositions should be performed from node *j* to node *i*. The initial truck allocation may cause costly empty truck movements. There are also some idle/inactive trucks which are awaiting at node *j* to the end of the planning horizon. This is because of the integer fleet size restrictions. In order to overcome this drawback, except the loaded truck positions, all the fleet size requirements, truck/trailer inventories, empty truck/trailer repositions, truck/trailer purchases and sales are represented by continuous decision variables.

Case-3: A comparatively better fleet allocation with relatively lesser empty truck repositions: 20 trucks are available at arc (*i, j*) and 14 trucks at arc (*j, i*) at the beginning of the planning horizon.

Truck inventory at Order Country (*i*)	**Trip-0** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
20 →	100	
Number of export positions		
	70 ←	14

Truck inventory at Order Country (*i*)	**Trip-1** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
14 →	80	
Number of export positions		
	56 ←	20

Truck inventory at Order Country (*i*)	**Trip-2** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
20 →	66	
Number of export positions		
	36 ←	14

Truck inventory at Order Country (*i*)	**Trip-3** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
14 →	46	0
Number of export positions		
0	22 ←	20

Truck inventory at Order Country (*i*)	**Trip-4** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
20 →	32	
Number of export positions		
	2 ←	14

Truck inventory at Order Country (*i*)	**Trip-5** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
Empty 10 + 2 →	12	0
Number of export positions		
0	0	2 **Awaiting**

Truck inventory at Order Country (*i*)	**Trip-6** Number of import positions	Truck inventory at Seaport in Turkey (*j*)
0	0	0
Number of export positions		
0	0	32 + 2 =34

Similar to the previous cases, all truck are concentrated at node *j* to the end of the planning horizon. The last case is comparatively better than the second case in terms of both the number of empty truck repositions and idle trucks.

The third model component incorporates relationships between truck/trailer inventories and relevant fleet sizing/allocation decisions. In the last model component, an empty vehicle repositioning concept can be divided into two categories, i.e. internal and external repositions (Baykasoğlu et al. 2015). In fact, these empty flows may be differentiated in terms of the source of repositioning, physical routes and repositioning times. Due to the imbalance of import/export loaded truck positions within the same time period, a need for internal empty truck/trailer repositioning occurs. An appropriate fleet allocation can provide minimal internal empty truck/trailer repositions. Furthermore, it is assumed throughout this paper that repositioning times are too short for the internal type empty flows. Thus, internal empty truck/trailer repositions can be realized within a monthly time period. On the other hand, due to the imbalance between the export truck/trailer fleet size and truck/trailer inventories in the consecutive time periods, a need for external empty truck/trailer repositioning arises. In detail, the difference between the outgoing truck fleet size from a node and the truck inventory of this node at the beginning of a time period can be satisfied by the external empty truck repositions. Contrary to internal empty truck/trailer repositioning, external empty truck/trailer repositions can be performed between any given two nodes where loaded truck/trailer movements are not possible. Furthermore, repositioning times are assumed to be one time period for the external type empty flows. In other words, external empty truck/trailer repositions realized at a time period *t* will satisfy the truck/trailer requirements of any given node in the subsequent time period, $t + 1$. Cabotage limitations should be considered while performing the external empty truck repositions with different types of plates.

The proposed intermodal fleet planning model is developed based on the following assumptions (the network notation and mathematical nomenclature with stochastic and fuzzy model parameters can be accessed at http://web.deu.edu.tr/baykasoglu/notation_fleet_planning_model.pdf):

- Outsourcing is only available for transportation operations between Turkey and order countries. It is not possible to outsource any truck or trailer for the transportation activities within Europe.
- The working days at order countries in Europe may differ from one period to another. The trucks and trailers are not available 100% of a time period.
- In contrast to the heterogeneous truck fleet which consists of European and Turkish plated trucks, the trailer fleet is assumed to be homogeneous.
- Truck requirements may occur at all nodes on the network whereas the trailer requirements can only occur at origins and destinations, i.e. seaports in Turkey and order countries.
- Empty truck/trailer repositions can only be performed by road freight transportation. In other words, empty trailers cannot be repositioned by means of Ro-Ro vessels and block trains.
- It is possible to reposition empty trucks without any trailer.
- It is not possible to reposition Turkish plated empty trucks without trailers in Europe due to cabotage limitations. This means that European plated empty trucks also cannot be repositioned in Turkey. On the other hand, Turkish plated trucks can be repositioned among the order countries in Europe on condition that moving with their own empty trailers to load extra freight and return to Turkey. On the other hand, there is no positional limitation on empty trailer repositioning in both Turkey and Europe.
- While determining fleet sizes, road-marine-rail transit times are only considered for loaded movements because empty truck/trailer repositioning times are already included in the fleet allocation decisions to satisfy loaded positions within a pre-specified time period.

The first goal of the proposed holistic intermodal fleet planning model is reformulated as in Equations (7)–(17). In fact, the road freight transportation cost is divided into three categories: costs for loaded trucks, empty trucks and trucks with empty trailers. The empty truck and trailer repositioning costs are defined for both internal and external empty flows. Moreover, fixed ownership costs of European/Turkish plated trucks and trailers are taken into account by the first objective function for both active and idle resources. Finally, purchasing costs for new trucks/trailers and revenues obtained from the sale of second-hand trucks/trailers are also included so as to determine periodic fleet expansion and reduction issues.

Goal-1: Minimize overall transport costs throughout the intermodal logistics network:

$$\begin{aligned}\text{Min} \quad \widetilde{Z1}_{Cost} \cong\ & \widetilde{Z1}_{marine} + \widetilde{Z1}_{railway} + \widetilde{Z1}_{road} + \widetilde{Z1}_{outsourcing} + \widetilde{Z1}_{ownership} \\ & + \widetilde{Z1}_{internal-reposition} + \widetilde{Z1}_{external-reposition} + \widetilde{Z1}_{idle} + \widetilde{Z1}_{purchasing} - \widetilde{Z1}_{sales}\end{aligned} \tag{7}$$

Marine transport costs by company-owned and other logistics service providers' Ro-Ro vessels:

$$\widetilde{Z1}_{marine} \cong \sum_{(i,j)\notin L_{ij}} \sum_{t\in T} R_{ijt}^{exp}.\widetilde{Co}_{ij}^{exp} + \sum_{(j,i)\notin L_{ji}} \sum_{t\in T} R_{jit}^{imp}.\widetilde{Co}_{ji}^{imp} + \sum_{(i,j)\notin C_{ij}} \sum_{t\in T} P_{ijt}^{exp}.\widetilde{Cp}_{ij}^{exp} + \sum_{(j,i)\notin C_{ji}} \sum_{t\in T} P_{jit}^{imp}.\widetilde{Cp}_{ji}^{imp} \tag{8}$$

Railway transport costs by the fixed schedule of block trains and flexible schedule of public trains:

$$\begin{aligned}\widetilde{Z1}_{railway} \cong & \sum_{(j,k)\notin L_{jk}} \sum_{t\in T} S_{jkt}^{exp}.\widetilde{Cb}_{jk}^{exp} + \sum_{(k,j)\notin L_{kj}} \sum_{t\in T} S_{kjt}^{imp}.\widetilde{Cb}_{kj}^{imp} + \sum_{(j,k)\notin C_{jk}} \sum_{t\in T} Pt_{jkt}^{exp}.\widetilde{Cp}_{jk}^{exp} \\ & + \sum_{(k,j)\notin C_{kj}} \sum_{t\in T} Pt_{kjt}^{imp}.\widetilde{Cp}_{kj}^{imp} + \sum_{k'\in T_k} \sum_{k^*\in G_k} \sum_{t\in T} Pt_{k^*k't}^{imp}.\widetilde{Cp}_{k^*k'}^{imp} \\ & + \sum_{k'\in T_k} \sum_{k^*\in G_k} \sum_{t\in T} Pt_{k'k^*t}^{exp}.\widetilde{Cp}_{k'k^*}^{exp}\end{aligned} \tag{9}$$

Road transport cost by European/Turkish plated trucks for loaded movements:

$$\begin{aligned}\widetilde{Z1}_{road} \cong & \sum_{j\in J} \sum_{l\in L} \sum_{t\in T} V_{jlt}^{exp}.d_{jl}^{r}.\widetilde{Cr} + \sum_{j\in J} \sum_{l\in L} \sum_{t\in T} V_{ljt}^{imp}.d_{lj}^{r}.\widetilde{Cr} + \sum_{k\in K} \sum_{l\in L} \sum_{t\in T} V_{klt}^{exp}.d_{kl}^{r}.\widetilde{Cr} \\ & + \sum_{k\in K} \sum_{l\in L} \sum_{t\in T} V_{lkt}^{imp}.d_{lk}^{r}.\widetilde{Cr} + \sum_{i\in I} \sum_{l\in L} \sum_{t\in T} (VE_{ilt}^{exp} + VT_{ilt}^{exp}).d_{il}^{r}.\widetilde{Cr} \\ & + \sum_{i\in I} \sum_{l\in L} \sum_{t\in T} (VE_{lit}^{imp} + VT_{lit}^{imp}).d_{li}^{r}.\widetilde{Cr}\end{aligned} \tag{10}$$

Outsourcing costs in the direct road and combined marine & road transportation:

$$\widetilde{Z1}_{outsource} \cong \sum_{i\in I} \sum_{l\in L} \sum_{t\in T} O_{ilt}^{exp}.\widetilde{SC}_{ilt}^{exp} + \sum_{i\in I} \sum_{l\in L} \sum_{t\in T} O_{lit}^{imp}.\widetilde{SC}_{lit}^{imp} \tag{11}$$

Fixed ownership costs for European/Turkish plated trucks and trailers:

$$\begin{aligned}\widetilde{Z1}_{ownership} \cong & \widetilde{OWC}_{Euro}^{Truck}.\left(\sum_{j\in J} \sum_{l\in L} \sum_{t\in T} (VS_{jlt}^{exp} + VS_{ljt}^{imp}) + \sum_{k\in K} \sum_{l\in L} \sum_{t\in T} (VS_{klt}^{exp} + VS_{lkt}^{imp}) \right. \\ & \left. + \sum_{i\in I} \sum_{l\in L} \sum_{t\in T} (EP_{ilt}^{exp} + EP_{lit}^{imp})\right) \\ & + \widetilde{OWC}_{Turk}^{Truck}.\left(\sum_{i\in I} \sum_{l\in L} \sum_{t\in T} (TP_{ilt}^{exp} + TP_{lit}^{imp})\right) \\ & + \widetilde{OWN}^{Trail}.\left(\sum_{i\in I} \sum_{l\in L} \sum_{t\in T} (TS_{ilt}^{exp} + TS_{lit}^{imp})\right)\end{aligned} \tag{12}$$

Internal empty truck and trailer repositioning costs:

$$\begin{aligned}\widetilde{Z1}_{int-reposition} \cong & \sum_{j\in J}\sum_{l\in L}\sum_{t\in T} V^{rep}_{jlt}.d^{r}_{jl}.\widetilde{RC}^{v} + \sum_{j\in J}\sum_{l\in L}\sum_{t\in T} V^{rep}_{ljt}.d^{r}_{lj}.\widetilde{RC}^{v} \\ & + \sum_{k\in K}\sum_{l\in L}\sum_{t\in T} V^{rep}_{klt}.d^{r}_{kl}.\widetilde{RC}^{v} + \sum_{k\in K}\sum_{l\in L}\sum_{t\in T} V^{rep}_{lkt}.d^{r}_{lk}.\widetilde{RC}^{v} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} (VE^{rep}_{ilt} + VT^{rep}_{ilt}).d^{r}_{il}.\widetilde{RC}^{v} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} (VE^{rep}_{lit} + VT^{rep}_{lit}).d^{r}_{li}.\widetilde{RC}^{v} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} TR^{rep}_{ilt}.d^{r}_{il}.\widetilde{RC}^{Tr} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} TR^{rep}_{lit}.d^{r}_{li}.\widetilde{RC}^{Tr} \end{aligned} \tag{13}$$

External empty truck and trailer repositioning costs:

$$\begin{aligned}\widetilde{Z1}_{ext-reposition} \cong & \sum_{k\in K}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{jkt}.d^{r}_{jk}.\widetilde{RC}^{v} + \sum_{k\in K}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{kjt}.d^{r}_{kj}.\widetilde{RC}^{v} \\ & + \sum_{j\in J}\sum_{j'\in J\setminus\{j\}}\sum_{t\in T} VR^{rep}_{j'jt}.d^{r}_{j'j}.\widetilde{RC}^{v} + \sum_{l\in L}\sum_{k\in K}\sum_{t\in T} VR^{rep}_{klt}.d^{r}_{kl}.\widetilde{RC}^{v} \\ & + \sum_{l\in L}\sum_{k\in K}\sum_{t\in T} VR^{rep}_{lkt}.d^{r}_{lk}.\widetilde{RC}^{v} + \sum_{k\in K}\sum_{k'\in K\setminus\{k\}}\sum_{t\in T} VR^{rep}_{k'kt}.d^{r}_{k'k}.\widetilde{RC}^{v} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} (VRE^{rep}_{ilt} + VRT^{rep}_{ilt}).d^{r}_{il}.\widetilde{RC}^{v} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} (VRE^{rep}_{lit} + VRT^{rep}_{lit}).d^{r}_{li}.\widetilde{RC}^{v} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} XTR^{rep}_{ilt}.d^{r}_{il}.\widetilde{RC}^{Tr} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} XTR^{rep}_{lit}.d^{r}_{li}.\widetilde{RC}^{Tr} + \sum_{i\in I}\sum_{i'\in I\setminus\{i\}}\sum_{t\in T} TR^{rep}_{i'it}.d^{r}_{i'i}.\widetilde{RC}^{Tr} \\ & + \sum_{l\in L}\sum_{l'\in L\setminus\{l\}}\sum_{t\in T} TR^{rep}_{l'lt}.d^{r}_{l'l}.\widetilde{RC}^{Tr} + \sum_{l\in L}\sum_{l'\in L\setminus\{l\}}\sum_{t\in T} TTR^{rep}_{l'lt}.d^{r}_{l'l}.\widetilde{RC}^{Tr} \end{aligned} \tag{14}$$

Fixed ownership costs for idle or inactive trucks and trailers:

$$\begin{aligned}\widetilde{Z1}_{idle} \cong & \widetilde{OWC}^{Truck}_{Euro}.\left(\sum_{j\in J}\sum_{t\in T} IVJ_{jt} + \sum_{k\in K}\sum_{t\in T} IVK_{kt} + \sum_{l\in L}\sum_{t\in T} IVL^{E}_{lt} + \sum_{i\in I}\sum_{t\in T} IVI^{E}_{it}\right) \\ & + \widetilde{OWC}^{Truck}_{Turk}.\left(\sum_{l\in L}\sum_{t\in T} IVL^{T}_{lt} + \sum_{i\in I}\sum_{t\in T} IVI^{T}_{it}\right) \\ & + \widetilde{OWN}^{Trail}.\left(\sum_{l\in L}\sum_{t\in T} ITL_{lt} + \sum_{i\in I}\sum_{t\in T} ITI_{it}\right) \end{aligned} \tag{15}$$

Purchasing costs for European/Turkish plated new trucks and trailers:

$$\widetilde{Z1}_{purchasing} \cong \sum_{i\in PR_I}\sum_{t\in T} Pr_{it}^{T}.\widetilde{PCV}^{T} + \sum_{l\in PR_L}\sum_{t\in T} Pr_{lt}^{E}.\widetilde{PCV}^{E} + \sum_{i\in PR_I}\sum_{t\in T} Prt_{it}.\widetilde{PCT} \quad (16)$$

Total revenue obtained from the sales of second-hand trucks and trailers:

$$\widetilde{Z1}_{sales} \cong \sum_{i\in PR_I}\sum_{t\in T} Sp_{it}^{T}.\widetilde{SV}^{T} + \sum_{l\in PR_L}\sum_{t\in T} Sp_{lt}^{E}.\widetilde{SV}^{E} + \sum_{i\in PR_I}\sum_{t\in T} Spt_{it}.\widetilde{ST} \quad (17)$$

The second goal of the proposed model can also be formulated as in the following Equations (18–24) by the inclusion of internal and external empty truck and trailer repositioning times.

Goal-2: Minimize total transit time including loading/unloading times, travel times and awaiting times at the terminals:

$$\begin{aligned}\text{Min}\quad \widetilde{Z2}_{Transit-time} \cong{}& \widetilde{Z2}_{direct-road} + \widetilde{Z2}_{combined-marine} + \widetilde{Z2}_{intermodal} + \widetilde{Z2}_{outsourcing} \\ &+ \widetilde{Z2}_{internal-reposition} + \widetilde{Z2}_{external-reposition}\end{aligned} \quad (18)$$

Road freight transit times for loaded truck movements and combined marine/road transit times:

$$\widetilde{Z2}_{direct-road} \cong \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} X_{ilt}^{exp}.\tilde{T}_{il}^{r} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} X_{lit}^{imp}.\tilde{T}_{li}^{r} \quad (19)$$

$$\begin{aligned}\widetilde{Z2}_{combined-marine} \cong{}& \sum_{i\in I}\sum_{j\in J}\sum_{l\in L}\sum_{t\in T} Y_{ijlt}^{exp}.(\tilde{T}_{ij}^{m} + \tilde{T}_{jl}^{r}) \\ &+ \sum_{i\in I}\sum_{j\in J}\sum_{l\in L}\sum_{t\in T} Y_{ljit}^{imp}.(\tilde{T}_{lj}^{r} + \tilde{T}_{ji}^{m})\end{aligned} \quad (20)$$

Combined marine, rail and road transit times in intermodal transportation:

$$\begin{aligned}\widetilde{Z2}_{intermodal} \cong{}& \sum_{i\in I}\sum_{j\in J}\sum_{k\notin G_k}\sum_{l\in L}\sum_{t\in T} Z_{ijklt}^{exp}.(\tilde{T}_{ij}^{m} + \tilde{T}_{jk}^{t} + \tilde{T}_{kl}^{r}) \\ &+ \sum_{i\in I}\sum_{j\in J}\sum_{k\notin G_k}\sum_{l\in L}\sum_{t\in T} Z_{lkjit}^{imp}.(\tilde{T}_{lk}^{r} + \tilde{T}_{kj}^{t} + \tilde{T}_{ji}^{m}) \\ &+ \sum_{i\in I}\sum_{j\in J}\sum_{k'\in T_k}\sum_{k^*\in G_k}\sum_{l\in G_l}\sum_{t\in T} Z_{ijk^*lt}^{exp}.(\tilde{T}_{ij}^{m} + \tilde{T}_{jk'}^{t} + \tilde{T}_{k'k^*}^{t} + \tilde{T}_{k^*l}^{r}) \\ &+ \sum_{i\in I}\sum_{j\in J}\sum_{k^*\in G_k}\sum_{k'\in T_k}\sum_{l\in G_l}\sum_{t\in T} Z_{lk^*jit}^{imp}.(\tilde{T}_{lk^*}^{r} + \tilde{T}_{k^*k'}^{t} + \tilde{T}_{k'j}^{t} + \tilde{T}_{ji}^{m})\end{aligned} \quad (21)$$

Marine and road freight transit times for outsourced transportation services:

$$\widetilde{Z2}_{outsourcing} \cong \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} O^{exp}_{ilt}.\left[\tilde{\varphi}^{Out}.\tilde{T}^{r}_{il} + (1-\tilde{\varphi}^{Out}).\sum_{j\in J}\tilde{T}^{m}_{ij} + \tilde{T}^{r}_{jl}\right] + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} O^{imp}_{lit}.\left[\tilde{\varphi}^{Out}.\tilde{T}^{r}_{li} + (1-\tilde{\varphi}^{Out}).\sum_{j\in J}\tilde{T}^{r}_{lj} + \tilde{T}^{m}_{ji}\right] \tag{22}$$

Road freight transit times for internal empty truck and trailer movements:

$$\begin{aligned}\widetilde{Z2}_{internal-reposition} \cong\ & \sum_{j\in J}\sum_{l\in L}\sum_{t\in T} V^{rep}_{jlt}.\tilde{T}^{r}_{jl} + \sum_{j\in J}\sum_{l\in L}\sum_{t\in T} V^{rep}_{ljt}.\tilde{T}^{r}_{lj} + \sum_{k\in K}\sum_{l\in L}\sum_{t\in T} V^{rep}_{klt}.\tilde{T}^{r}_{kl} \\ & + \sum_{k\in K}\sum_{l\in L}\sum_{t\in T} V^{rep}_{lkt}.\tilde{T}^{r}_{lk} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T}(VE^{rep}_{ilt} + VT^{rep}_{ilt}).\tilde{T}^{r}_{il} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} TR^{rep}_{ilt}.\tilde{T}^{r}_{il} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T}(VE^{rep}_{lit} + VT^{rep}_{lit}).\tilde{T}^{r}_{li} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} TR^{rep}_{lit}.\tilde{T}^{r}_{li}\end{aligned} \tag{23}$$

Road freight transit times for the external empty truck and trailer movements:

$$\begin{aligned}\widetilde{Z2}_{external-rep} \cong\ & \sum_{k\in K}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{jkt}.\tilde{T}^{r}_{jk} + \sum_{k\in K}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{kjt}.\tilde{T}^{r}_{kj} + \sum_{j\in J}\sum_{j'\in J\setminus\{j\}}\sum_{t\in T} VR^{rep}_{j'jt}.\tilde{T}^{r}_{j'j} \\ & + \sum_{l\in L}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{jlt}.\tilde{T}^{r}_{jl} + \sum_{l\in L}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{ljt}.\tilde{T}^{r}_{lj} + \sum_{l\in L}\sum_{l'\in L\setminus\{l\}}\sum_{t\in T} VR^{rep}_{l'lt}.\tilde{T}^{r}_{l'l} \\ & + \sum_{i\in I}\sum_{i'\in I\setminus\{i\}}\sum_{t\in T} VR^{rep}_{i'it}.\tilde{T}^{r}_{i'i} + \sum_{l\in L}\sum_{k\in K}\sum_{t\in T} VR^{rep}_{klt}.\tilde{T}^{r}_{kl} + \sum_{l\in L}\sum_{k\in K}\sum_{t\in T} VR^{rep}_{lkt}.\tilde{T}^{r}_{lk} \\ & + \sum_{k\in K}\sum_{k'\in K\setminus\{k\}}\sum_{t\in T} VR^{rep}_{k'kt}.\tilde{T}^{r}_{k'k} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T}(VRE^{rep}_{ilt} + VRT^{rep}_{ilt}).\tilde{T}^{r}_{il} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T}(VRE^{rep}_{lit} + VRT^{rep}_{lit}).\tilde{T}^{r}_{li} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} XTR^{rep}_{ilt}.d_{il}.\tilde{T}^{r}_{il} \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} XTR^{rep}_{lit}.\tilde{T}^{r}_{li} + \sum_{i\in I}\sum_{i'\in I\setminus\{i\}}\sum_{t\in T} TR^{rep}_{i'it}.\tilde{T}^{r}_{i'i} \\ & + \sum_{l\in L}\sum_{l'\in L\setminus\{l\}}\sum_{t\in T} TR^{rep}_{l'lt}.\tilde{T}^{r}_{l'l} + \sum_{l\in L}\sum_{l'\in L\setminus\{l\}}\sum_{t\in T} TTR^{rep}_{l'lt}.\tilde{T}^{r}_{l'l}\end{aligned} \tag{24}$$

The last goal, which provides environmenally conscious fleet plans, is also formulated as follows by incorporating CO_2 emission values during loaded and empty truck/trailer repositions.

Goal-3: Minimize total environmental impact (CO_2 emissions) by different transport modes:

$$\begin{aligned}\text{Min}\quad \tilde{Z}_{CO2-emission} \cong\ & \widetilde{Z3}_{road} + \widetilde{Z3}_{marine} + \widetilde{Z3}_{railway} + \widetilde{Z3}_{outsourcing} \\ & + \widetilde{Z3}_{internal-reposition} + \widetilde{Z3}_{external-reposition}\end{aligned} \tag{25}$$

Total CO_2 emissions from the loaded truck movements by road freight transport:

$$\begin{aligned}\widetilde{Z3}_{road} \cong & \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} X^{exp}_{ilt}.d^{r}_{il}.\widetilde{CO2}^{R} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} X^{imp}_{lit}.d^{r}_{li}.\widetilde{CO2}^{R} \\ & + \sum_{i\in I}\sum_{j\in J}\sum_{l\in L}\sum_{t\in T} Y^{imp}_{ljit}.d^{r}_{lj}.\widetilde{CO2}^{R} + \sum_{i\in I}\sum_{j\in J}\sum_{l\in L}\sum_{t\in T} Y^{exp}_{ijlt}.d^{r}_{jl}.\widetilde{CO2}^{R} \\ & + \sum_{i\in I}\sum_{j\in J}\sum_{k^*\in G_k}\sum_{l\in G_l}\sum_{t\in T} Z^{imp}_{lk^*jit}.d^{r}_{lk^*}.\widetilde{CO2}^{R} \end{aligned} \tag{26}$$

Total CO_2 emissions from the maritime logistics services:

$$\begin{aligned}\widetilde{Z3}_{marine} \cong & \sum_{(i,j)\notin L_{ij}}\sum_{t\in T} R^{exp}_{ijt}.d^{m}_{ij}.\widetilde{CO2}^{M} + \sum_{(j,i)\notin L_{ji}}\sum_{t\in T} R^{imp}_{jit}.d^{m}_{ji}.\widetilde{CO2}^{M} \\ & + \sum_{(i,j)\notin C_{ij}}\sum_{t\in T} P^{exp}_{ijt}.d^{m}_{ij}.\widetilde{CO2}^{M} + \sum_{(j,i)\notin C_{ji}}\sum_{t\in T} P^{imp}_{jit}.d^{m}_{ji}.\widetilde{CO2}^{M} \end{aligned} \tag{27}$$

Total CO_2 emissions from railway transport services:

$$\begin{aligned}\widetilde{Z3}_{railway} \cong & \sum_{(j,k)\notin L_{jk}}\sum_{t\in T} S^{exp}_{jkt}.\widetilde{Cap}^{exp}_{jk}.d^{t}_{jk}.\widetilde{CO2}^{T} + \sum_{(k,j)\notin L_{kj}}\sum_{t\in T} S^{imp}_{kjt}.\widetilde{Cap}^{imp}_{kj}.d^{t}_{kj}.\widetilde{CO2}^{T} \\ & + \sum_{(j,k)\notin C_{jk}}\sum_{t\in T} Pt^{exp}_{jkt}.d^{t}_{jk}.\widetilde{CO2}^{T} \sum_{(k,j)\notin C_{kj}}\sum_{t\in T} Pt^{imp}_{kjt}.d^{t}_{kj}.\widetilde{CO2}^{T} \\ & + \sum_{k'\in T_k}\sum_{k^*\in G_k}\sum_{t\in T} Pt^{exp}_{k'k^*t}.d^{t}_{k'k^*}.\widetilde{CO2}^{T} + \sum_{k'\in T_k}\sum_{k^*\in G_k}\sum_{t\in T} Pt^{imp}_{k^*k't}.d^{t}_{k^*k'}.\widetilde{CO2}^{T} \end{aligned} \tag{28}$$

Total CO_2 emissions from maritime logistics and road freight transport of outsourced services:

$$\begin{aligned}\widetilde{Z3}_{outsourcing} \cong & \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} O^{exp}_{ilt}.\left[\tilde{\varphi}^{Out}.d^{r}_{il}.\widetilde{CO2}^{R} + (1-\tilde{\varphi}^{Out}).\sum_{j\in J} d^{m}_{ij}.\widetilde{CO2}^{M} + d^{r}_{jl}.\widetilde{CO2}^{R}\right] \\ & + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} O^{imp}_{lit}.\left[\tilde{\varphi}^{Out}.d^{r}_{li}.\widetilde{CO2}^{R} + (1-\tilde{\varphi}^{Out}).\sum_{j\in J} d^{r}_{lj}.\widetilde{CO2}^{R} + d^{m}_{ji}.\widetilde{CO2}^{M}\right] \end{aligned} \tag{29}$$

Total CO_2 emissions from internal empty truck and trailer movements in road

transportation:

$$\widetilde{Z3}_{internal-reposition} \cong \sum_{j\in J}\sum_{l\in L}\sum_{t\in T} V^{rep}_{ljt}.d^{r}_{lj}.\widetilde{CO2}^{RE} + \sum_{j\in J}\sum_{l\in L}\sum_{t\in T} V^{rep}_{jlt}.d^{r}_{jl}.\widetilde{CO2}^{RE} + \sum_{k\in K}\sum_{l\in L}\sum_{t\in T} V^{rep}_{klt}.d^{r}_{kl}.\widetilde{CO2}^{RE} + \sum_{k\in K}\sum_{l\in L}\sum_{t\in T} V^{rep}_{lkt}.d^{r}_{lk}.\widetilde{CO2}^{RE} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} (VE^{rep}_{ilt} + VT^{rep}_{ilt}).d^{r}_{il}.\widetilde{CO2}^{RE} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} (VE^{rep}_{lit} + VT^{rep}_{lit}).d^{r}_{li}.\widetilde{CO2}^{RE} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} TR^{rep}_{ilt}.d^{r}_{il}.\widetilde{CO2}^{R} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} TR^{rep}_{lit}.d^{r}_{li}.\widetilde{CO2}^{R} \quad (30)$$

Total CO_2 emissions from external empty truck and trailer movements in road transportation:

$$\widetilde{Z3}_{external-reposition} \cong \sum_{k\in K}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{jkt}.d^{r}_{jk}.\widetilde{CO2}^{RE} + \sum_{k\in K}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{kjt}.d^{r}_{kj}.\widetilde{CO2}^{RE} + \sum_{j\in J}\sum_{j'\in J\setminus\{j\}}\sum_{t\in T} VR^{rep}_{j'jt}.d^{r}_{j'j}.\widetilde{CO2}^{RE} + \sum_{l\in L}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{jlt}.d^{r}_{jl}.\widetilde{CO2}^{RE} + \sum_{l\in L}\sum_{j\in J}\sum_{t\in T} VR^{rep}_{ljt}.d^{r}_{lj}.\widetilde{CO2}^{RE} + \sum_{l\in L}\sum_{l'\in L\setminus\{l\}}\sum_{t\in T} VR^{rep}_{l'lt}.d^{r}_{l'l}.\widetilde{CO2}^{RE} + \sum_{i\in I}\sum_{i'\in I\setminus\{i\}}\sum_{t\in T} VR^{rep}_{i'it}.d^{r}_{i'i}.\widetilde{CO2}^{RE} + \sum_{l\in L}\sum_{k\in K}\sum_{t\in T} VR^{rep}_{klt}.d^{r}_{kl}.\widetilde{CO2}^{RE} + \sum_{l\in L}\sum_{k\in K}\sum_{t\in T} VR^{rep}_{lkt}.d^{r}_{lk}.\widetilde{CO2}^{RE} + \sum_{k\in K}\sum_{k'\in K\setminus\{k\}}\sum_{t\in T} VR^{rep}_{k'kt}.d^{r}_{k'k}.\widetilde{CO2}^{RE} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} (VRE^{rep}_{ilt} + VRT^{rep}_{ilt}).d^{r}_{il}.\widetilde{CO2}^{RE} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} (VRE^{rep}_{lit} + VRT^{rep}_{lit}).d^{r}_{li}.\widetilde{CO2}^{RE} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} XTR^{rep}_{ilt}.d^{r}_{il}.\widetilde{CO2}^{R} + \sum_{i\in I}\sum_{l\in L}\sum_{t\in T} XTR^{rep}_{lit}.d^{r}_{li}.\widetilde{CO2}^{R} + \sum_{i\in I}\sum_{i'\in I\setminus\{i\}}\sum_{t\in T} TR^{rep}_{i'it}.d^{r}_{i'i}.\widetilde{CO2}^{R} + \sum_{l\in L}\sum_{l'\in L\setminus\{l\}}\sum_{t\in T} TR^{rep}_{l'lt}.d^{r}_{l'l}.\widetilde{CO2}^{R} + \sum_{l\in L}\sum_{l'\in L\setminus\{l\}}\sum_{t\in T} TTR^{rep}_{l'lt}.d^{r}_{l'l}.\widetilde{CO2}^{R} \quad (31)$$

Model component I: Multi-mode load planning or intermodal freight transportation planning:

(i) *Periodic load allocation to different modes of transport in order to satisfy probabilistic import/export transport demands of order countries*:

$$Pr\left\{ X_{ilt}^{exp} + \sum_{j\in J} Y_{ijlt}^{exp} + \sum_{j\in D_j}\sum_{k\in K} Z_{ijklt}^{exp} + O_{ilt}^{exp} \geq De_{ilt}^{exp} \right\} \geq \alpha_{ilt} \quad (32)$$
$$\forall i \in I,\ \forall l \in L,\ \forall t \in T$$

$$Pr\left\{ X_{lit}^{imp} + \sum_{j\in J} Y_{ljit}^{imp} + \sum_{j\in D_j}\sum_{k\in K} Z_{lkjit}^{imp} + O_{lit}^{imp} \geq De_{lit}^{imp} \right\} \geq \beta_{lit} \quad (33)$$
$$\forall i \in I, \forall l \in L, \forall t \in T$$

(ii) *The specific constraints on marine transportation services:*

$$\sum_{l\in L} Y_{ijlt}^{exp} \leq \tilde{N}_{ij^*t}^{exp}.\widetilde{Cap}_{ij^*}^{exp} - \sum_{l\in L} Y_{ij^*lt}^{exp} - \sum_{l\in L}\sum_{k\in K} Z_{ij^*klt}^{exp} \quad (34)$$
$$\forall (i,j) \in C_{ij} \ \wedge\ j \in T_j\ ,\ j^* \in \gamma_j, \forall t \in T$$

$$\sum_{l\in L} Y_{ljit}^{imp} \leq \tilde{N}_{j^*it}^{imp}.\widetilde{Cap}_{j^*i}^{imp} - \sum_{l\in L} Y_{lj^*it}^{imp} - \sum_{l\in L}\sum_{k\in K} Z_{llkj^*it}^{imp} \quad (35)$$
$$\forall (j,i) \in C_{ji} \ \wedge\ j \in T_j\ ,\ j^* \in \gamma_j, \forall t \in T$$

$$\sum_{l\in L} Y_{ijlt}^{exp} \leq \tilde{N}_{ijt}^{exp}.\widetilde{Cap}_{ij}^{exp} \quad \forall (i,j) \in C_{ij} \ \wedge\ j \notin D_j, \forall t \in T \quad (36)$$

$$\sum_{l\in L} Y_{ljit}^{imp} \leq \tilde{N}_{jit}^{imp}.\widetilde{Cap}_{ji}^{imp} \quad \forall (j,i) \in C_{ji} \ \wedge\ j \notin D_j, \forall t \in T \quad (37)$$

$$\sum_{l\in L} Y_{ijlt}^{exp} \leq \tilde{N}_{ijt}^{exp}.\widetilde{Cap}_{ij}^{exp} + P_{ijt}^{exp} \quad \forall (i,j) \in B_{ij} \ \wedge\ j \notin D_j,\ \forall t \in T \quad (38)$$

$$\sum_{l\in L} Y_{ljit}^{imp} \leq \tilde{N}_{jit}^{imp}.\widetilde{Cap}_{ji}^{imp} + P_{jit}^{imp} \quad \forall (j,i) \in B_{ji} \ \wedge\ j \notin D_j,\ \forall t \in T \quad (39)$$

$$\sum_{l\in L} Y_{ijlt}^{exp} = R_{ijt}^{exp} \quad \{\forall (i,j) \in C_{ij} \vee B_{ij}\} \wedge \{\, j \in T_j \wedge\ j \notin D_j\},\ \forall t \in T \quad (40)$$

$$\sum_{l\in L} Y_{ljit}^{imp} = R_{jit}^{imp} \quad \{\forall (j,i) \in C_{ji} \vee B_{ji}\} \wedge \{\, j \in T_j \wedge\ j \notin D_j\},\ \forall t \in T \quad (41)$$

$$\sum_{l\in L} Y^{exp}_{ijlt} + \sum_{k\in K}\sum_{l\in L} Z^{exp}_{ijklt} \le \tilde{N}^{exp}_{ijt}.\widetilde{Cap}^{exp}_{ij} - \sum_{l\in L} Y^{exp}_{ij'lt} + P^{exp}_{ijt} \quad \forall (i,j)\in C_{ij},\ j\in D_j \wedge j\in \gamma_j,\ j'\in T_j, \forall t\in T \tag{42}$$

$$\sum_{l\in L} Y^{imp}_{ljit} + \sum_{k\in K}\sum_{l\in l} Z^{imp}_{lkjit} \le \tilde{N}^{imp}_{jit}.\widetilde{Cap}^{imp}_{ji} - \sum_{l\in L} Y^{imp}_{lj'it} + P^{imp}_{jit} \quad \forall (j,i)\in C_{ji}, j\in D_j \wedge j\in \gamma_j,\ j'\in T_j, \forall t\in T \tag{43}$$

$$\sum_{l\in L} Y^{exp}_{ijlt} + \sum_{k\in K}\sum_{l\in L} Z^{exp}_{ijklt} = R^{exp}_{ijt} + P^{exp}_{ijt} \quad \forall (i,j)\in B_{ij}\vee C_{ij},\ j\in D_j, \forall t\in T \tag{44}$$

$$\sum_{l\in L} Y^{imp}_{ljit} + \sum_{k\in K}\sum_{l\in l} Z^{imp}_{lkjit} = R^{imp}_{jit} + P^{imp}_{jit} \quad \forall (j,i)\in B_{ji}\vee C_{ji},\ j\in D_j, \forall t\in T \tag{45}$$

$$\sum_{l\in L} Y^{exp}_{ijlt} + \sum_{k\in K}\sum_{l\in L} Z^{exp}_{ijklt} \le P^{exp}_{ijt} \quad \forall (i,j)\notin C_{ij},\ j\in D_j, \forall t\in T \tag{46}$$

$$\sum_{l\in L} Y^{imp}_{ljit} + \sum_{k\in K}\sum_{l\in L} Z^{imp}_{lkjit} \le P^{imp}_{jit} \quad \forall (j,i)\notin C_{ji},\ j\in D_j, \forall t\in T \tag{47}$$

$$\sum_{l\in L} Y^{exp}_{ijlt} \le P^{exp}_{ijt} \quad \forall (i,j)\in L_{ij},\ j\notin D_j, \forall t\in T \tag{48}$$

$$\sum_{l\in L} Y^{imp}_{ljit} \le P^{imp}_{jit} \quad \forall (j,i)\in L_{ji},\ j\notin D_j \forall t\in T \tag{49}$$

$$R^{exp}_{ijt}/\widetilde{Cap}^{exp}_{ij} = MS^{exp}_{ijt} \quad \forall (i,j)\notin L_{ij},\ j\notin \gamma_j, \forall t\in T \tag{50}$$

$$R^{imp}_{jit}/\widetilde{Cap}^{imp}_{ji} = MS^{imp}_{jit} \quad \forall (j,i)\notin L_{ji},\ j\notin \gamma_j, \forall t\in T \tag{51}$$

$$(R^{exp}_{ijt} + R^{exp}_{ij^*t})/\widetilde{Cap}^{exp}_{ij^*} = MS^{exp}_{ij^*t} \quad \forall (i,j)\notin L_{ij},\ j\in T_j,\ j^*\in \gamma_j, \forall t\in T \tag{52}$$

$$(R^{imp}_{jit} + R^{imp}_{j^*it})/\widetilde{Cap}^{imp}_{j^*i} = MS^{imp}_{j^*it} \quad \forall (j,i)\notin L_{ji}, j\in T_j,\ j^*\in \gamma_j, \forall t\in T \tag{53}$$

$$U^{exp}_{ijt} = R^{exp}_{ijt}/(\tilde{N}^{exp}_{ijt}.\widetilde{Cap}^{exp}_{ij}) \quad \forall (i,j)\notin L_{ij}, \forall t\in T \tag{54}$$

$$U^{imp}_{jit} = R^{imp}_{jit}/(\tilde{N}^{imp}_{jit}.\widetilde{Cap}^{imp}_{ji}) \quad \forall (j,i)\notin L_{ji}, \forall t\in T \tag{55}$$

$$\widetilde{LB}^{exp}_{ijt} \le U^{exp}_{ijt} \ \ and \ \ \widetilde{LB}^{imp}_{jit} \le U^{imp}_{jit} \quad \forall (i,j)\notin L_{ij}, \forall t\in T, \forall (j,i)\notin L_{ji} \tag{56}$$

The specific constraints on railway transportation services:

$$\sum_{i\in I}\sum_{l\in L} Z^{exp}_{ijklt} \le S^{exp}_{jkt}.Cap^{exp}_{jk} \quad (j,k)\in C_{jk}\backslash\{T_k\}, \forall \in T \tag{57}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{imp}_{lkjit} \le S^{imp}_{kjt}.Cap^{imp}_{kj} \quad (k,j)\in C_{kj}\backslash\{T_k\}, \forall \in T \tag{58}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{exp}_{ijklt} \le Pt^{exp}_{jkt} \quad (j,k)\in L_{jk}, \forall t \in T \tag{59}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{imp}_{lkjit} \le Pt^{imp}_{kjt} \quad (k,j)\in L_{kj}, \forall t \in T \tag{60}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{exp}_{ijklt} \le S^{exp}_{jkt}.Cap^{exp}_{jk} + Pt^{exp}_{jkt} \quad (j,k)\in B_{jk}\backslash\{T_k\}, \forall t \in T \tag{61}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{imp}_{lkjit} \le S^{imp}_{kjt}.Cap^{imp}_{kj} + Pt^{imp}_{kjt} \quad (k,j)\in B_{kj}\backslash\{T_k\}, \forall t \in T \tag{62}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{exp}_{ijk'lt} + \sum_{i\in I}\sum_{l\in G_l}\sum_{k^*\in G_k} Z^{exp}_{ijk^*lt} \le S^{exp}_{jk't}.Cap^{exp}_{jk'} \quad \forall j\in D_j, \forall k'\in T_k \cap^{(} j,k')\in C_{jk'}, \forall t\in T \tag{63}$$

$$\sum_{i\in I}\sum_{l\in L} Z^{imp}_{lk'jit} + \sum_{i\in I}\sum_{l\in G_l}\sum_{k^*\in G_k} Z^{imp}_{lk^*jit} \le S^{imp}_{k'jt}.Cap^{imp}_{k'j} \quad \forall j\in D_j, \forall k'\in T_k\cap(k',j)\in C_{k'j}, \forall t\in T \tag{64}$$

$$\sum_{i\in I}\sum_{j\in D_j}\sum_{l\in G_l} Z^{exp}_{ijk^*lt} \le Pt^{exp}_{k'k^*t} \quad \forall k'\in T_k, \forall k^*\in G_k\cap(k',k^*)\in L_{k'k^*}, \forall t\in T \tag{65}$$

$$\sum_{i\in I}\sum_{j\in D_j}\sum_{l\in G_l} Z^{imp}_{lk^*jit} \le Pt^{imp}_{k^*k't} \quad \forall k'\in T_k, \forall k\in G_k\cap(k^*,k')\in L_{k^*k'}, \forall t\in T \tag{66}$$

$$\widetilde{LB}^{exp}_{jkt} \le S^{exp}_{jkt} \le \widetilde{UB}^{exp}_{jkt} \ \ and \ \ \widetilde{LB}^{imp}_{kjt} \le S^{imp}_{kjt} \le \widetilde{UB}^{imp}_{kjt} \quad \forall(k,j)\notin L_{kj}, \ \forall(j,k)\notin L_{jk}, \forall t\in T \tag{67}$$

$$|S^{exp}_{jkt} - S^{imp}_{kjt}| \le 1 \quad \forall(k,j)\notin L_{kj}\ , \forall(j,k)\notin L_{jk}, \forall t\in T \tag{68}$$

$$\left|\sum_{t\in T}(S^{exp}_{jkt} - S^{imp}_{kjt})\right| \le 1 \quad \forall(k,j)\notin L_{kj}\ , \forall(j,k)\notin L_{jk} \tag{69}$$

The detailed explanations related to the constraints of the multi-mode load planning component of the proposed intermodal fleet planning model are given in Baykasoğlu and Subulan (2016). The import/export transport demands of the order countries cannot be known with certainty but have known probability distributions. Therefore, demand satisfaction constraints (32) and (33) are formulated as linear probabilistic restrictions with stochastic right-hand sides. Furthermore, all the cost coefficients, transit times,

transport capacities of Ro-Ro vessels, maximum number of marine cruises, lower and upper bounds on block train services are considered as unknown parameters and represented by triangular fuzzy numbers.

Model Component II: Truck and trailer fleet size optimization and fleet allocation:

The constraints of truck/trailer fleet sizing, fleet allocation, loaded truck movements and internal empty truck/trailer repositioning issues can be formulated as in Equations (70–102):

(iv) *Connections between load plans and loaded truck movements:*

$$X_{ilt}^{exp} = VE_{ilt}^{exp} + VT_{ilt}^{exp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{70}$$

$$\sum_{i \in I} Y_{ijlt}^{exp} = V_{jlt}^{exp} \quad \forall j \in J,\ \forall l \in L, \forall t \in T \tag{71}$$

$$\sum_{i \in I} \sum_{j \in J} Z_{ijklt}^{exp} = V_{klt}^{exp} \quad \forall k \in K,\ \forall l \in L, \forall t \in T \tag{72}$$

$$X_{lit}^{imp} = VE_{lit}^{imp} + VT_{lit}^{imp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{73}$$

$$\sum_{i \in I} Y_{ljit}^{imp} = V_{ljt}^{imp} \quad \forall j \in J,\ \forall l \in L, \forall t \in T \tag{74}$$

$$\sum_{i \in I} \sum_{j \in J} Z_{lkjit}^{imp} = V_{lkt}^{imp} \quad \forall k \in K,\ \forall l \in L, \forall t \in T \tag{75}$$

(v) *The minimum number of European and Turkish plated truck fleet sizes:*

$$V_{jlt}^{exp}.\tilde{T}_{jl}^{r} + V_{ljt}^{imp}.\tilde{T}_{lj}^{r} + |V_{jlt}^{exp}.\tilde{T}_{jl}^{r} - V_{ljt}^{imp}.\tilde{T}_{lj}^{r}| \le (VS_{jlt}^{exp} + VS_{ljt}^{imp}).\tilde{\varphi}_{lt}.\Gamma.\tilde{\delta}^{E} \quad \forall j \in J,\ \forall l \in L, \forall t \in T \tag{76}$$

$$V_{klt}^{exp}.\tilde{T}_{kl}^{r} + V_{lkt}^{imp}.\tilde{T}_{lk}^{r} + |V_{klt}^{exp}.\tilde{T}_{kl}^{r} - V_{lkt}^{imp}.\tilde{T}_{lk}^{r}| \le (VS_{klt}^{exp} + VS_{lkt}^{imp}).\tilde{\varphi}_{lt}.\Gamma.\tilde{\delta}^{E} \quad \forall k \in K,\ \forall l \in L, \forall t \in T \tag{77}$$

$$VE_{ilt}^{exp}.\tilde{T}_{il}^{r} + VE_{lit}^{imp}.\tilde{T}_{li}^{r} + |VE_{ilt}^{exp}.\tilde{T}_{il}^{r} - VE_{lit}^{imp}.\tilde{T}_{li}^{r}| \le (EP_{ilt}^{exp} + EP_{lit}^{imp}).\tilde{\varphi}_{lt}.\Gamma.\tilde{\delta}^{E} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{78}$$

$$VT_{ilt}^{exp}.\tilde{T}_{il}^{r} + VT_{lit}^{imp}.\tilde{T}_{li}^{r} + |VT_{ilt}^{exp}.\tilde{T}_{il}^{r} - VT_{lit}^{imp}.\tilde{T}_{li}^{r}| \le (TP_{ilt}^{exp} + TP_{lit}^{imp}).\tilde{\varphi}_{lt}.\Gamma.\tilde{\delta}^{T} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{79}$$

(vi) *Logical constraints for the relationship between loaded truck movements & fleet sizes*:

$$VS_{jlt}^{exp} \le V_{jlt}^{exp} \quad \forall j \in J,\ \forall l \in L, \forall t \in T \tag{80}$$

$$VS_{ljt}^{imp} \le V_{ljt}^{imp} \quad \forall j \in J,\ \forall l \in L, \forall t \in T \tag{81}$$

$$VS_{klt}^{exp} \le V_{klt}^{exp} \quad \forall k \in K,\ \forall l \in L, \forall t \in T \tag{82}$$

$$VS_{lkt}^{imp} \le V_{lkt}^{imp} \quad \forall k \in K,\ \forall l \in L, \forall t \in T \tag{83}$$

$$EP_{ilt}^{exp} \le VE_{ilt}^{exp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{84}$$

$$EP_{lit}^{imp} \le VE_{lit}^{imp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{85}$$

$$TP_{ilt}^{exp} \le VT_{ilt}^{exp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{86}$$

$$TP_{lit}^{imp} \le VT_{lit}^{imp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{87}$$

(vii) *The minimum number of trailer fleet size on the overall intermodal logistics network*:

$$\begin{aligned} & X_{ilt}^{exp}.\tilde{T}_{li}^{r} + \sum_{j\in J} Y_{ijlt}^{exp}.(\tilde{T}_{ij}^{m} + \tilde{T}_{jl}^{r}) + \sum_{j\in D_j}\sum_{k\in K} Z_{ijklt}^{exp}.(\tilde{T}_{ij}^{m} + \tilde{T}_{jk}^{t} + \tilde{T}_{kl}^{r}) + X_{lit}^{imp}.\tilde{T}_{li}^{r} \\ & + \sum_{j\in J} Y_{ljit}^{imp}.(\tilde{T}_{ji}^{m} + \tilde{T}_{lj}^{r}) + \sum_{j\in D_j}\sum_{k\in K} Z_{lkjit}^{imp}.(\tilde{T}_{ji}^{m} + \tilde{T}_{kj}^{t} + \tilde{T}_{lk}^{r}) \\ & + \Bigg| X_{ilt}^{exp}.\tilde{T}_{il} + \sum_{j\in J} Y_{ijlt}^{exp}.(\tilde{T}_{ij}^{m} + \tilde{T}_{jl}^{r}) \sum_{j\in D_j}\sum_{k\in K} Z_{ijklt}^{exp}.(\tilde{T}_{ij}^{m} + \tilde{T}_{jk}^{t} + \tilde{T}_{kl}^{r}) - X_{lit}^{imp}.\tilde{T}_{li}^{r} \\ & - \sum_{j\in J} Y_{ljit}^{imp}.(\tilde{T}_{ji}^{m} + \tilde{T}_{lj}^{r}) - \sum_{j\in D_j}\sum_{k\in K} Z_{lkjit}^{imp}.(\tilde{T}_{ji}^{m} + \tilde{T}_{kj}^{t} + \tilde{T}_{lk}^{r}) \Bigg| \\ & \le (TS_{ilt}^{exp} + TS_{lit}^{imp}).\tilde{\varphi}_{lt}.\tilde{\Gamma}.\tilde{\varepsilon} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \end{aligned} \tag{88}$$

(viii) *Logical constraints for the relationship between loaded trailer movements & fleet sizes*:

$$TS_{ilt}^{exp} \le X_{ilt}^{exp} + \sum_{j\in J} Y_{ijlt}^{exp} + \sum_{j\in J}\sum_{k\in K} Z_{ijklt}^{exp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{89}$$

$$TS_{lit}^{imp} \le X_{lit}^{imp} + \sum_{j\in J} Y_{ljit}^{imp} + \sum_{j\in J}\sum_{k\in K} Z_{lkjit}^{imp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{90}$$

(ix) *The internal empty truck repositioning constraints*:

$$V_{ljt}^{rep} \ge V_{jlt}^{exp} - V_{ljt}^{imp} - VS_{jlt}^{exp} \quad \forall j \in J,\ \forall l \in L, \forall t \in T \tag{91}$$

$$V_{jlt}^{rep} \ge V_{ljt}^{imp} - V_{jlt}^{exp} - VS_{ljt}^{imp} \quad \forall j \in J,\ \forall l \in L, \forall t \in T \tag{92}$$

$$V_{lkt}^{rep} \ge V_{klt}^{exp} - V_{lkt}^{imp} - VS_{klt}^{exp} \quad \forall k \in K,\ \forall l \in L, \forall t \in T \tag{93}$$

$$V_{klt}^{rep} \ge V_{lkt}^{imp} - V_{klt}^{exp} - VS_{lkt}^{imp} \quad \forall k \in K,\ \forall l \in L, \forall t \in T \tag{94}$$

$$VE_{lit}^{rep} \ge VE_{ilt}^{exp} - VE_{lit}^{imp} - EP_{ilt}^{exp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{95}$$

$$VE_{ilt}^{rep} \ge VE_{lit}^{imp} - VE_{ilt}^{exp} - EP_{lit}^{imp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{96}$$

$$VT_{lit}^{rep} \geq VT_{ilt}^{exp} - VT_{lit}^{imp} - TP_{ilt}^{exp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{97}$$

$$VT_{ilt}^{rep} \geq VT_{lit}^{imp} - VT_{ilt}^{exp} - TP_{lit}^{imp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{98}$$

(x) *The internal empty trailer repositioning constraints*:

$$TR_{lit}^{rep} \geq X_{ilt}^{exp} + \sum_{j\in J} Y_{ijlt}^{exp} + \sum_{j\in J}\sum_{k\in K} Z_{ijklt}^{exp} - X_{lit}^{imp} - \sum_{j\in J} Y_{ljit}^{imp} - \sum_{j\in J}\sum_{k\in K} Z_{lkjit}^{imp} - TS_{ilt}^{exp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{99}$$

$$TR_{ilt}^{rep} \geq X_{lit}^{imp} + \sum_{j\in J} Y_{ljit}^{imp} + \sum_{j\in J}\sum_{k\in K} Z_{lkjit}^{imp} - X_{ilt}^{exp} - \sum_{j\in J} Y_{ijlt}^{exp} - \sum_{j\in J}\sum_{k\in K} Z_{ijklt}^{exp} - TS_{lit}^{imp} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{100}$$

(xi) *Logical constraints for the relationship between internal empty truck/trailer repositioning*:

$$VE_{ilt}^{rep} + VT_{ilt}^{rep} \geq TR_{ilt}^{rep} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{101}$$

$$VE_{lit}^{rep} + VT_{lit}^{rep} \geq TR_{lit}^{rep} \quad \forall i \in I,\ \forall l \in L, \forall t \in T \tag{102}$$

Constraints (70–75) provide connections between load plans and loaded truck movements. In other words, the amount of loaded truck positions between different locations can be computed by these constraints. By making use of constraints (76–79), the minimum number of the truck fleet size that should be available on each road transport route can be calculated. In detail, truck fleet size on different transport routes are computed with the help of loaded truck positions, road transit times and the number of working days, truck/trailer availabilities and length of the planning period (List et al., 2003). The minimum number of European plated truck fleet size between each railway station and order countries can be obtained from constraint set (77). Additionally, truck fleet sizes are determined by considering the uncertainty of road transit times, truck availabilities and number of working days. Constraints (80–87) ensure that the number of trucks assigned to any route on the road transport network should be less than or equal to the number of loaded truck positions on the same route. In reality, these are the logical constraints for the relationship between loaded truck positions and fleet size requirements. The trailer fleet size on the logistics network can be calculated by using constraint (88). In addition to the road transit times, marine and rail transit times are also incorporated while determining the trailer fleet size. Because, trailers can be transferred by Ro-Ro vessels and block or public trains without their trucks on marine and railway networks. On the contrary to truck fleet sizes, we can just calculate the trailer fleet size between the seaports in Turkey and order countries. Similar to constraints (80–87), constraints (89–90) ensure that the number of import/export trailers assigned to any route cannot exceed the number of loaded trailer positions on the same route. Constraints (91–98) are related to the internal empty truck repositioning problem of the transportation company. In fact, these repositions result from the difference between the import/export loaded truck positions of any given two nodes within the same time period. For instance,

constraints (97) and (98) present Turkish plated internal empty truck repositioning between seaports in Turkey and order countries. For performing the internal empty truck repositions between any given two nodes, it is a necessary condition that loaded truck positions must be possible between these nodes. Both European and Turkish plated trucks can be internally repositioned between seaports in Turkey and order countries. In summary, the required number of internal empty truck repositions for any given node can be calculated by subtracting the incoming loaded truck positions and truck fleet size of this node from the outgoing loaded truck positions. According to constraints (99) and (100), the internal empty trailer repositions between the seaports in Turkey and order countries can be calculated. It is assumed that empty trailers can be transferred by using only road freight transportation. Constraints (101) and (102) ensure that the amount of empty trucks repositioned between the seaports in Turkey and order countries should be greater than the number of empty trailer repositions. In other words, sufficient number of empty trucks should be available for empty trailer movements.

Model Component - III: Periodic truck and trailer inventories on the intermodal logistics network and fleet expansion/reduction decisions:

The constraints of periodic truck/trailer inventories, idle/inactive trucks/trailers and fleet expansion/reduction can be formulated as presented by the following Equations (103–121):

(xii) *The* periodic *European & Turkish plated truck inventories at different locations:*

$$\begin{aligned} VJ_{jt} = VJ_{jt-1} &+ \sum_{l\in L} V^{imp}_{ljt} - \sum_{l\in L} V^{exp}_{jlt} + \sum_{l\in L} V^{rep}_{ljt} - \sum_{l\in L} V^{rep}_{jlt} + \sum_{j'\neq j} VR^{rep}_{j'jt} \\ &- \sum_{j'\neq j} VR^{rep}_{jj't} + \sum_{k\in K} VR^{rep}_{kjt} - \sum_{k\in K} VR^{rep}_{jkt} + \sum_{l\in L} VR^{rep}_{ljt} \\ &- \sum_{l\in L} VR^{rep}_{jlt} \quad \forall j \in J,\ \forall t \in T \end{aligned} \tag{103}$$

$$\begin{aligned} VK_{kt} = VK_{kt-1} &+ \sum_{l\in L} V^{imp}_{lkt} - \sum_{l\in L} V^{exp}_{klt} + \sum_{l\in L} V^{rep}_{lkt} - \sum_{l\in L} V^{rep}_{klt} + \sum_{k'\neq k} VR^{rep}_{k'kt} \\ &- \sum_{k'\neq k} VR^{rep}_{kk't} + \sum_{j\in J} VR^{rep}_{jkt} - \sum_{j\in J} VR^{rep}_{kjt} + \sum_{l\in L} VR^{rep}_{lkt} \\ &- \sum_{l\in L} VR^{rep}_{klt} \quad \forall k \in K,\ \forall t \in T \end{aligned} \tag{104}$$

$$\begin{aligned} VL^{E}_{lt} = VL^{E}_{lt-1} &+ Pr^{E}_{lt} - Sp^{E}_{lt} + \sum_{j\in J} V^{exp}_{jlt} - \sum_{j\in J} V^{imp}_{ljt} + \sum_{k\in K} V^{exp}_{klt} - \sum_{k\in K} V^{imp}_{lkt} + \sum_{i\in I} VE^{exp}_{ilt} \\ &- \sum_{i\in I} VE^{imp}_{lit} - \sum_{j\in J} V^{rep}_{ljt} + \sum_{j\in J} V^{rep}_{jlt} - \sum_{k\in K} V^{rep}_{lkt} + \sum_{k\in K} V^{rep}_{klt} - \sum_{i\in I} VE^{rep}_{lit} \\ &+ \sum_{i\in I} VE^{rep}_{ilt} - \sum_{j\in J} VR^{rep}_{ljt} + \sum_{j\in J} VR^{rep}_{jlt} - \sum_{k\in K} VR^{rep}_{lkt} + \sum_{k\in K} VR^{rep}_{klt} - \sum_{l'\neq l} VR^{rep}_{ll't} \\ &+ \sum_{l'\neq l} VR^{rep}_{l'lt} - \sum_{i\in I} VRE^{rep}_{lit} + \sum_{i\in I} VRE^{rep}_{ilt} \quad \forall l \in PR_L, \forall t \in T \end{aligned} \tag{105}$$

$$VL_{lt}^{T} = VL_{lt-1}^{T} + \sum_{i\in I} VT_{ilt}^{exp} - \sum_{i\in I} VT_{lit}^{imp} + \sum_{i\in I} VT_{ilt}^{rep} - \sum_{i\in I} VT_{lit}^{rep} + \sum_{l'\neq l} TTR_{l'lt}^{rep} - \sum_{l'\neq l} TTR_{ll't}^{rep} - \sum_{i\in I} VRT_{lit}^{rep} + \sum_{i\in I} VRT_{ilt}^{rep} \quad \forall l \in L, \forall t \in T \tag{106}$$

$$VI_{it}^{E} = VI_{it-1}^{E} + \sum_{l\in L} VE_{lit}^{imp} - \sum_{l\in L} VE_{ilt}^{exp} + \sum_{l\in L} VE_{lit}^{rep} - \sum_{l\in L} VE_{ilt}^{rep} + \sum_{l\in L} VRE_{lit}^{rep} - \sum_{l\in L} VRE_{ilt}^{rep} \quad \forall i \in I,\ \forall t \in T \tag{107}$$

$$VI_{it}^{T} = VI_{it-1}^{T} + Pr_{it}^{T} - Sp_{it}^{T} + \sum_{l\in L} VT_{lit}^{imp} - \sum_{l\in L} VT_{ilt}^{exp} + \sum_{l\in L} VT_{lit}^{rep} - \sum_{l\in L} VT_{ilt}^{rep} + \sum_{l\in L} VRT_{lit}^{rep} - \sum_{l\in L} VRT_{ilt}^{rep} + \sum_{i'\neq i} VRT_{i'it}^{rep} - \sum_{i'\neq i} VRT_{ii't}^{rep} \quad \forall i \in PR_I,\ \forall t \in T \tag{108}$$

(xiii) *Periodic trailer inventories at seaports in Turkey & order countries in Europe*:

$$TI_{it} = TI_{it-1} + Prt_{it} - Spt_{it} + \sum_{l\in L} X_{lit}^{imp} + \sum_{l\in L}\sum_{j\in J} Y_{ljit}^{imp} + \sum_{l\in L}\sum_{j\in J}\sum_{k\in K} Z_{lkjit}^{imp} - \sum_{l\in L} X_{ilt}^{exp} - \sum_{l\in L}\sum_{j\in J} Y_{ijlt}^{exp} - \sum_{l\in L}\sum_{j\in J}\sum_{k\in K} Z_{ijklt}^{exp} + \sum_{l\in L} TR_{lit}^{rep} - \sum_{l\in L} TR_{ilt}^{rep} + \sum_{l\in L} XTR_{lit}^{rep} - \sum_{l\in L} XTR_{ilt}^{rep} + \sum_{i'\neq i} TR_{i'it}^{rep} - \sum_{i'\neq i} TR_{ii't}^{rep} \quad \forall i \in PR_I,\ \forall t \in T \tag{109}$$

$$TL_{lt} = TL_{lt-1} + \sum_{i\in I} X_{ilt}^{exp} + \sum_{i\in I}\sum_{j\in J} Y_{ijlt}^{exp} + \sum_{i\in I}\sum_{j\in J}\sum_{k\in K} Z_{ijklt}^{exp} - \sum_{i\in I} X_{lit}^{imp} - \sum_{i\in I}\sum_{j\in J} Y_{ljit}^{imp} - \sum_{i\in I}\sum_{j\in J}\sum_{k\in K} Z_{lkjit}^{imp} + \sum_{i\in I} TR_{ilt}^{rep} - \sum_{i\in I} TR_{lit}^{rep} + \sum_{i\in I} XTR_{ilt}^{rep} - \sum_{i\in I} XTR_{lit}^{rep} + \sum_{l'\neq l} TR_{l'lt}^{rep} - \sum_{l'\neq l} TR_{ll't}^{rep} + \sum_{l'\neq l} TTR_{l'lt}^{rep} - \sum_{l'\neq l} TTR_{ll't}^{rep} \quad \forall l \in L,\ \forall t \in T \tag{110}$$

(xiv) Logical *constraints for relationship between truck/trailer inventory levels & fleet sizes:*

$$\sum_{j\in J}\sum_{l\in L} VS^{exp}_{jlt} + \sum_{j\in J}\sum_{l\in L} VS^{imp}_{ljt} + \sum_{k\in K}\sum_{l\in L} VS^{exp}_{klt} + \sum_{k\in K}\sum_{l\in L} VS^{imp}_{lkt} + \sum_{i\in I}\sum_{l\in L} EP^{exp}_{ilt} + \sum_{i\in I}\sum_{l\in L} EP^{imp}_{lit} \le \sum_{j\in J} VJ_{jt-1} + \sum_{k\in K} VK_{kt-1} + \sum_{i\in I} VI^{E}_{it-1} + \sum_{l\in L} VL^{E}_{lt-1} \quad \forall t \in T \tag{111}$$

$$\sum_{i\in I}\sum_{l\in L} TP^{exp}_{ilt} + \sum_{i\in I}\sum_{l\in L} TP^{imp}_{lit} \le \sum_{i\in I} VI^{T}_{it-1} + \sum_{l\in L} VL^{T}_{lt-1} \quad \forall t \in T \tag{112}$$

$$\sum_{i\in I}\sum_{l\in L} TS^{exp}_{ilt} + \sum_{i\in I}\sum_{l\in L} TS^{imp}_{lit} \le \sum_{i\in I} TI_{it-1} + \sum_{l\in L} TL_{lt-1} \quad \forall t \in T \tag{113}$$

(xv) *The idle trucks & trailers awaiting at different locations on the logistics network*:

$$IVJ_{jt} = VJ_{jt-1} - \sum_{l\in L} VS^{exp}_{jlt} \quad \forall j \in J,\ \forall t \in T \tag{114}$$

$$IVK_{kt} = VK_{kt-1} - \sum_{l\in L} VS^{exp}_{klt} \quad \forall k \in K,\ \forall t \in T \tag{115}$$

$$IVL^{E}_{lt} = VL^{E}_{lt-1} - \sum_{j\in J} VS^{imp}_{ljt} - \sum_{k\in K} VS^{imp}_{lkt} - \sum_{i\in I} EP^{imp}_{lit} \quad \forall l \in L, \forall t \in T \tag{116}$$

$$IVL^{T}_{lt} = VL^{T}_{lt-1} - \sum_{i\in I} TP^{imp}_{lit} \quad \forall l \in L, \forall t \in T \tag{117}$$

$$IVI^{E}_{it} = VI^{E}_{it-1} - \sum_{l\in L} EP^{exp}_{ilt} \quad \forall i \in I, \forall t \in T \tag{118}$$

$$IVI^{T}_{it} = VI^{T}_{it-1} - \sum_{l\in L} TP^{exp}_{ilt} \quad \forall i \in I, \forall t \in T \tag{119}$$

$$ITL_{lt} = TL_{lt-1} - \sum_{i\in I} TS^{imp}_{lit} \quad \forall l \in L, \forall t \in T \tag{120}$$

$$ITI_{it} = TI_{it-1} - \sum_{l\in L} TS^{exp}_{ilt} \quad \forall i \in I, \forall t \in T \tag{121}$$

Constraints (103–108) present periodic European/Turkish plated truck inventories at different locations. In detail, (103–105) are formulated to calculate the periodic European plated truck inventories at seaports in Europe, railway stations and order countries. In order to compute the truck inventories of any given node at the end of a time period, initial truck inventory levels, outgoing loaded truck positions, incoming loaded truck positions, internal and external empty truck repositions, new truck purchases and sales of second-hand trucks are taken into consideration. It is assumed that newly purchased European plated trucks or second-hand trucks can only join or leave from the network in Germany. Constraint (106) is designated for periodic Turkish plated truck inventories

at the order countries. Both European and Turkish plated truck inventories are hold at the seaports in Turkey as in Equations (107 and 108). On the other hand, it is assumed that newly purchased Turkish plated trucks or second-hand trucks can join or leave the system at Istanbul only. Constraint (109) reflects periodic trailer inventory levels at the seaports in Turkey. It should be emphasized here that there are no trailer inventories at seaports in Europe and rail stations since they are directly transferred from/to these terminals to order countries or seaports in Turkey. Finally, the monthly trailer inventory levels at the order countries can be computed via constraint set (110). The relationship between the truck and trailer inventories and their fleet sizes are provided by constraints (111–113). These constraints maintain that the number of active trucks/trailers cannot exceed total inventory levels of trucks/trailers at the end of the previous time period. For instance, as formulated in constraint (111), the number of European plated trucks utilized in time period *t*, cannot exceed the total inventory levels at seaports in Turkey and Europe, railway stations and order countries at the end of the previous time period, *t*-1. Similar restrictions are written for Turkish plated trucks/trailers in constraints (112) and (113). The number of idle trucks awaiting at the seaports, railway stations and order countries can be computed by constraints (114–121). These idle trucks can be calculated by subtracting the required fleet sizes at any given time period from the beginning inventory levels of the nodes. Furthermore, truck fleet size can be utilized for realizing both loaded truck positions and internal empty truck repositions. Constraints (120) and (121) are formulated for calculating the number of idle or inactive trailers.

Model Component - IV: External empty truck and trailer repositioning issues:

The external empty truck and trailer repositioning constraints and their relationships can be given as in the following Equations (122–141):

(xvi) *The* external *European & Turkish plated empty truck repositions:*

$$VJ_{j0} \geq \sum_{l \in L} VS_{jlt}^{exp} \quad \forall j \in J,\ t = 1 \tag{122}$$

$$\sum_{j' \neq j} VR_{j'jt-1}^{rep} + \sum_{k \in K} VR_{kjt-1}^{rep} + \sum_{l \in L} VR_{ljt-1}^{rep} \geq \sum_{l \in L} VS_{jlt}^{exp} - VJ_{jt-1} \quad \forall j \in J,\ t > 1 \tag{123}$$

$$VK_{k0} \geq \sum_{l \in L} VS_{klt}^{exp} \quad \forall k \in K,\ t = 1 \tag{124}$$

$$\sum_{k' \neq k} VR_{k'kt-1}^{rep} + \sum_{j \in J} VR_{jkt-1}^{rep} + \sum_{l \in L} VR_{lkt-1}^{rep} \geq \sum_{l \in L} VS_{klt}^{exp} - VK_{kt-1} \quad \forall k \in K,\ t > 1 \tag{125}$$

$$VL_{l0}^{E} \geq \sum_{j \in J} VS_{ljt}^{imp} + \sum_{k \in K} VS_{lkt}^{imp} + \sum_{i \in I} EP_{lit}^{imp} \quad \forall l \in L,\ t = 1 \tag{126}$$

$$\sum_{j\in J} VR^{rep}_{jlt-1} + \sum_{k\in K} VR^{rep}_{klt-1} + \sum_{l'\neq l} VR^{rep}_{l'lt-1} + \sum_{i\in I} VRE^{rep}_{ilt-1} \geq \sum_{j\in J} VS^{imp}_{ljt} + \sum_{k\in K} VS^{imp}_{lkt} + \sum_{i\in I} EP^{imp}_{lit} - VL^{E}_{lt-1} \quad \forall l \in L,\ t > 1 \tag{127}$$

$$VL^{T}_{l0} \geq \sum_{i\in I} TP^{imp}_{lit} \quad \forall l \in L,\ \ t = 1 \tag{128}$$

$$\sum_{l'\neq l} TTR^{rep}_{l'lt-1} + \sum_{i\in I} VRT^{rep}_{ilt-1} \geq \sum_{i\in I} TP^{imp}_{lit} - VL^{T}_{lt-1} \quad \forall l \in L,\ t > 1 \tag{129}$$

$$VI^{E}_{i0} \geq \sum_{l\in L} EP^{exp}_{ilt} \quad \forall i \in I,\ \ t = 1 \tag{130}$$

$$\sum_{l\in L} VRE^{rep}_{lit-1} \geq \sum_{l\in L} EP^{exp}_{ilt} - VI^{E}_{it-1} \quad \forall i \in I,\ \ t > 1 \tag{131}$$

$$VI^{T}_{i0} \geq \sum_{l\in L} TP^{exp}_{ilt} \quad \forall i \in I,\ \ t = 1 \tag{132}$$

$$\sum_{l\in L} VRT^{rep}_{lit-1} + \sum_{i'\neq i} VRT^{rep}_{i'it-1} \geq \sum_{l\in L} TP^{exp}_{ilt} - VI^{T}_{it-1} \quad \forall i \in I,\ t > 1 \tag{133}$$

(xvii) *The* external *empty trailer repositions:*

$$TL_{l0} \geq \sum_{i\in I} TS^{imp}_{lit} \quad \forall l \in L,\ \ t = 1 \tag{134}$$

$$\sum_{i\in I} XTR^{rep}_{ilt-1} + \sum_{l'\neq l} TR^{rep}_{l'lt-1} + \sum_{l'\neq l} TTR^{rep}_{l'lt-1} \geq \sum_{i\in I} TS^{imp}_{lit} - TL_{lt-1} \quad \forall l \in L,\ t > 1 \tag{135}$$

$$TI_{i0} \geq \sum_{l\in L} TS^{exp}_{ilt} \quad \forall i \in I,\ \ t = 1 \tag{136}$$

$$\sum_{l\in L} XTR^{rep}_{lit-1} + \sum_{i'\neq i} TR^{rep}_{i'it-1} \geq \sum_{l\in L} TS^{exp}_{ilt} - TI_{it-1} \quad \forall i \in I,\ t > 1 \tag{137}$$

The logical constraints for relationship between external empty truck & trailer repositions:

$$VR^{rep}_{ll't} \geq TR^{rep}_{ll't} \quad \forall l \neq l' \in L,\ \forall t \in T \tag{138}$$

$$VRT^{rep}_{ii't} \geq TR^{rep}_{ii't} \quad \forall i \neq i' \in I,\ \forall t \in T \tag{139}$$

$$VRE^{rep}_{ilt} + VRT^{rep}_{ilt} \geq XTR^{rep}_{ilt} \quad \forall i \in I, \forall l \in L, \forall t \in T \tag{140}$$

$$VRE^{rep}_{lit} + VRT^{rep}_{lit} \geq XTR^{rep}_{lit} \quad \forall i \in I, \forall l \in L, \forall t \in T \tag{141}$$

(xviii) *Non-negativity and integrality restrictions of decision variables:*

$$0 \leq U_{ijt}^{exp} \leq 1 \quad \text{and} \quad 0 \leq U_{jit}^{imp} \leq 1 \quad \forall (i,j) \notin L_{ij}, \forall (j,i) \notin L_{ji}, \forall t \in T \tag{142}$$

$$\begin{aligned} &X_{ilt}^{exp},\ X_{lit}^{imp},\ Y_{ijlt}^{exp},\ Y_{ljit}^{imp},\ Z_{ijklt}^{exp},\ Z_{lkjit}^{imp}, O_{ilt}^{exp}, O_{ilt}^{imp}, P_{ijt}^{exp} \\ &P_{jit}^{imp}, Pt_{jkt}^{exp}, Pt_{kjt}^{imp}, S_{jkt}^{exp}, S_{kjt}^{imp},\ Pt_{k'k^*t}^{exp}, Pt_{k^*k't}^{imp}, V_{jlt}^{exp} \\ &V_{klt}^{exp}, V_{ljt}^{imp}, V_{lkt}^{imp}, VE_{lit}^{imp}, VE_{ilt}^{exp}, VT_{lit}^{imp}, VT_{ilt}^{exp} \geq 0 \text{ and integer.} \end{aligned} \tag{143}$$

Constraints (122–133) are related to external empty truck repositions. The difference between the outgoing fleet size of a node and truck inventory of this node at the beginning of a time period can be satisfied by the external empty truck repositions of the previous time period. If the required number of outgoing trucks (fleet size) of any given node at a time period are greater than the current truck inventory of that node, external truck repositions from other locations to the relevant node are needed. It is assumed that external empty truck repositions of the previous time period can satisfy the truck requirements of the present time period. So, a sufficient number of trucks should be available at the beginning of the planning horizon so as to use them for loaded truck positions. Constraints (122) and (123) are formulated for the external European plated empty truck repositions to the seaports in Europe. Constraint (125) ensures that external empty truck repositions can be performed from the other railway stations, seaports in Europe and order countries to a railway station. According to constraint (127), European plated empty trucks can be repositioned from the seaports in Turkey and Europe, railway stations and other order countries to any given order country. Between the order countries and seaports in Turkey, both European and Turkish plated empty trucks can be repositioned. Constraint (129) maintains that Turkish plated empty trucks can be transferred from the seaports in Turkey and other order countries to any given order country. Besides, Turkish plated trucks are generally repositioned with their empty trailers among the order countries. Constraints (130–133) deal with the external European and Turkish plated empty truck repositions to seaports in Turkey. In a similar way, constraints (134–137) are derived for external empty trailer repositions. The external empty trailer repositions can be realized among the order countries themselves, among the seaports themselves in Turkey and also between the seaports in Turkey and order countries. It should be mentioned here that, unlike the internal empty truck and trailer repositioning, it is possible to realize external repositions on the routes which exclude the loaded truck/trailer movements. Constraints (138–141) present the relationship between external empty truck and trailer repositions. In reality, there should be a satisfactory amount of empty trucks so as to realize the empty trailer flows. Constraint (138) ensures that empty trailer repositions among the order countries themselves can be performed by using European plated trucks. Similarly, it is warranted by constraint (139) that empty trailers can be repositioned among the seaports themselves in Turkey via Turkish plated trucks. In practice, these constraints are derived based on the cabotage limitations. Finally, constraint (142) defines the lower and upper bounds for decision variables related to the utilization of company-owned Ro-Ro vessels and constraint (143) preserves the non-negativity of the decision variables and maintains that some variables must take integer values.

3. A hybrid fuzzy and stochastic programming approach

In this section, a hybrid fuzzy and stochastic programming approach is applied to the proposed intermodal fleet planning model for a better reflection of the real-world applications. In fact, most real-world fleet planning problems in intermodal transportation may include different types of uncertainties such as randomness, fuzziness and dynamism simultaneously. In this research, a hybrid chance-constrained stochastic programming method and a well-known fuzzy linear programming approach are employed simultaneously in order to convert the fuzzy and stochastic model into an equivalent auxiliary crisp form.

3.1. Chance-constrained stochastic programming application

It is supposed that the monthly import/export freight demands of the order countries may not be known with certainty but may have known probability distributions. Given this variability in freight demand, it may not always be practical to satisfy all of the transportation requirements 100% of the time. Instead, it is targeted to find a fleet plan which will satisfy the probabilistic freight demand constraints under a pre-specified percentage, i.e. 90 or 95% of the time. It makes sure that generated fleet plans will be feasible for 95% of all instances of the random freight demands. The deterministic equivalent form of these probabilistic constraints can be obtained by computing the critical values of the random right-hand sides. The critical value can be determined by integrating density function and inverse cumulative distribution function of the random variable (Bisschop and Roelofs 2006). This paper maintains that total amounts of loads transferred by different modes of transport should meet the random freight demands which are represented by independent continuous random variables with cumulative distributions. The freight demands of the order countries are assumed to fit a normal distribution with known means and standard deviations as follows:

$$De_{ilt}^{exp} \sim N(\mu_{ilt}^{exp}, (\sigma_{ilt}^{exp})^2) \quad \forall i \in I,\ \forall l \in L,\ \forall t \in T \tag{144}$$

$$De_{lit}^{imp} \sim N(\mu_{lit}^{imp}, (\sigma_{lit}^{imp})^2) \quad \forall i \in I,\ \forall l \in L,\ \forall t \in T \tag{145}$$

The deterministic approximation of the probabilistic demand constraints can be formulated as follows by using the inverse cumulative or quantile function of the normal distribution:

$$X_{ilt}^{exp} + \sum_{j\in J} Y_{ijlt}^{exp} + \sum_{j\in D_j}\sum_{k\in K} Z_{ijklt}^{exp} + O_{ilt}^{exp} \geq \zeta_{ilt} \quad \forall i \in I,\ \forall l \in L,\ \forall t \in T \tag{146}$$

$$\text{Cumulative distribution function:} F(\zeta_{ilt}) = \alpha_{ilt} \quad \forall i \in I,\ \forall l \in L,\ \forall t \in T \tag{147}$$

$$X_{lit}^{imp} + \sum_{j\in J} Y_{ljit}^{imp} + \sum_{j\in D_j}\sum_{k\in K} Z_{lkjit}^{imp} + O_{lit}^{imp} \geq \psi_{lit} \quad \forall i \in I,\ \forall l \in L,\ \forall t \in T \tag{148}$$

$$\text{Cumulative distribution function:} F(\psi_{lit}) = \beta_{lit} \quad \forall i \in I,\ \forall l \in L,\ \forall t \in T \tag{149}$$

$$\zeta_{ilt} = F^{-1}(\alpha_{ilt}|\mu_{ilt}^{exp}, \sigma_{ilt}^{exp}) = \mu_{ilt}^{exp} + \sigma_{ilt}^{exp}.\sqrt{2}.\ erf^{-1}(2.\alpha_{ilt} - 1) \quad \forall i \in I,\ \forall l \in L,\ \forall t \in T, \tag{150}$$

$$\psi_{lit} = F^{-1}(\beta_{lit}|\mu_{lit}^{imp}, \sigma_{lit}^{imp}) = \mu_{lit}^{imp} + \sigma_{lit}^{imp}.\sqrt{2}.\ erf^{-1}(2.\beta_{lit} - 1) \quad \forall i \in I,\ \forall l \in L,\ \forall t \in T \tag{151}$$

where *erf* represents the so-called error function. The chance-constrained stochastic programming model was constructed using some private functions in LINGO 15.0 optimization software. For instance, the @SPSTGRNDV function is utilized for identifying random variables. Additionally, the @SPDISTNORM function is used for declaring a normal distribution to random freight demands. Moreover, the @SPSAMPSIZE function is employed to set sample size. Finally, in order to assign probabilistic constraints to chance constraint sets, the @SPCHANCE function is needed.

3.2. Application of a resolution method to the fuzzy intermodal fleet planning model

In this section, the interactive resolution method of Jimenez et al. (2007) for linear programming models with fuzzy parameters are selected to transform the proposed fuzzy intermodal fleet planning model into the crisp equivalent form. Through the fuzzy intermodal fleet planning model, various alternative fleet plans can be generated with different degrees of satisfaction. In summary, there is a conflict among the satisfaction degree of goals and fulfilment degree of constraints, so α-acceptable optimal solutions can be achieved. The most important reason for choosing the aforementioned interactive resolution method is that a computationally efficient way can be achieved of solving possibilistic programming problems with its linearity and without any increment in the objective functions and inequality constraints. Furthermore, this method is able to provide feasible solutions with various degrees of feasibility (the α-parametric approach). Moreover, the decision maker may also have additional information about the risk of violation related to the model constraints. Finally, by adopting the resolution method of Jimenez et al. (2007), the following auxiliary crisp equivalent goals and constraints are obtained. For the sake of simplicity, only the first cost component of total transport cost objective (9) and some possibilistic constraints such as (36, 53, 56 and 76) are transformed and then non-fuzzy constraints are added to the fuzzy model with the remaining unaltered.

Transformation of the fuzzy objective function coefficients: uncertain cost terms using their expected value:

$$\begin{aligned} Min \quad EV(Z1_{marine}) = & \sum_{(i,j)\notin L_{ij}} \sum_{t\in T} R_{ijt}^{exp} \cdot \left(\frac{Co_{ij}^{exp/pes} + 2.Co_{ij}^{exp/mos} + Co_{ij}^{exp/opt}}{4} \right) \\ & + \sum_{(j,i)\notin L_{ji}} \sum_{t\in T} R_{jit}^{imp} \cdot \left(\frac{Co_{ji}^{\overline{imp}/pes} + 2.Co_{ji}^{\overline{imp}/mos} + Co_{ji}^{\overline{imp}/opt}}{4} \right) \\ & + \sum_{(i,j)\notin C_{ij}} \sum_{t\in T} P_{ijt}^{exp} \cdot \left(\frac{Cp_{ij}^{exp/pes} + 2.Cp_{ij}^{exp/mos} + Cp_{ij}^{exp/opt}}{4} \right) \\ & + \sum_{(j,i)\notin C_{ji}} \sum_{t\in T} P_{jit}^{imp} \cdot \left(\frac{Cp_{ji}^{imp/pes} + 2.Cp_{ji}^{imp/mos} + Cp_{ji}^{imp/opt}}{4} \right) \end{aligned} \tag{152}$$

Transformation of fuzzy technological coefficients and right-hand side values of constraints by their expected interval and fuzzy ranking procedure of Jimenez (1996):

$$\sum_{l\in L} Y_{ijlt}^{exp} \le \alpha\left(\frac{N_{ijt}^{exp/pes}+N_{ijt}^{exp/mos}}{2}\right)\left(\frac{Cap_{ij}^{exp/pes}+Cap_{ij}^{exp/mos}}{2}\right) + (1-\alpha)\left(\frac{N_{ijt}^{exp/mos}+N_{ijt}^{exp/opt}}{2}\right)\left(\frac{Cap_{ij}^{exp/mos}+Cap_{ij}^{exp/opt}}{2}\right)$$
$$\forall (i,j)\in C_{ij} \ \wedge j \notin D_j, \forall t \in T \tag{153}$$

$$R_{jit}^{imp} + R_{j^*it}^{imp} \le \frac{\alpha}{2}\left(\frac{Cap_{ji}^{imp/pes}+Cap_{ji}^{imp/mos}}{2}\right)MS_{j^*it}^{imp} + \left(1-\frac{\alpha}{2}\right) \times \left(\frac{Cap_{ji}^{imp/mos}+Cap_{ji}^{imp/opt}}{2}\right)MS_{j^*it}^{imp} \tag{154}$$

$$R_{jit}^{imp} + R_{j^*it}^{imp} \ge \frac{\alpha}{2}\left(\frac{Cap_{ji}^{imp/mos}+Cap_{ji}^{imp/opt}}{2}\right).MS_{j^*it}^{imp} + \left(1-\frac{\alpha}{2}\right)\left(\frac{Cap_{ji}^{imp/pes}+Cap_{ji}^{imp/mos}}{2}\right).MS_{j^*it}^{imp}$$
$$\forall (j,i)\notin L_{ji}, j\in T_j\ ,\ j^*\in \gamma_j, \forall t\in T \tag{155}$$

$$U_{jit}^{imp} \ge \alpha\left(\frac{LB_{jit}^{imp/mos}+LB_{jit}^{imp/opt}}{2}\right) + (1-\alpha)\left(\frac{LB_{jit}^{imp/pes}+LB_{jit}^{imp/mos}}{2}\right) \tag{156}$$
$$\forall (j,i)\notin L_{ji},\ \forall t\in T$$

$$V_{jlt}^{exp}.\left\{\alpha\left(\frac{T_{jl}^{r/mos}+T_{jl}^{r/opt}}{2}\right)+(1-\alpha)\left(\frac{T_{jl}^{r/pes}+T_{jl}^{r/mos}}{2}\right)\right\} + V_{ljt}^{imp}.\left\{\alpha\left(\frac{T_{lj}^{r/mos}+T_{lj}^{r/opt}}{2}\right)+(1-\alpha)\left(\frac{T_{lj}^{r/pes}+T_{lj}^{r/mos}}{2}\right)\right\} + \left|V_{jlt}^{exp}.\left\{\alpha\left(\frac{T_{jl}^{r/mos}+T_{jl}^{r/opt}}{2}\right)+(1-\alpha)\left(\frac{T_{jl}^{r/pes}+T_{jl}^{r/mos}}{2}\right)\right\} - V_{ljt}^{imp}.\left\{\alpha\left(\frac{T_{lj}^{r/mos}+T_{lj}^{r/opt}}{2}\right)+(1-\alpha)\left(\frac{T_{lj}^{r/pes}+T_{lj}^{r/mos}}{2}\right)\right\}\right|$$
$$\le \left(VS_{jlt}^{exp}+VS_{ljt}^{imp}\right).\Gamma.\left\{\alpha\left(\frac{\varphi_{lt}^{pes}+\varphi_{lt}^{mos}}{2}\right)\left(\frac{\delta^{E/pes}+\delta^{E/mos}}{2}\right)+(1-\alpha)\left(\frac{\varphi_{lt}^{mos}+\varphi_{lt}^{opt}}{2}\right)\left(\frac{\delta^{E/mos}+\delta^{E/opt}}{2}\right)\right\}$$
$$\forall j\in J,\ \forall l\in L, \forall t\in T \tag{157}$$

where α is a parameter which indicates the feasibility degree of constraints ($\alpha = 0$ unacceptable or most risky solution, $\alpha = 0.5$ neither acceptable nor unacceptable solution, $\alpha = 1.0$ completely acceptable or risk-free solution, etc.).

4. A real-world case study of an international transportation company

4.1. Company introduction and data description

In this section, the performance of the proposed holistic intermodal fleet planning model is illustrated through a real-world application in a large-sized international transportation company in Turkey. The intermodal transportation network configuration of this application is demonstrated in Figure 5. In addition to Turkish employees, the company has approximately 1239 employees in 12 different countries across Europe. As mentioned previously, the transportation company performs international transportation services all over Europe. Thus, it seeks to reduce on-road mileage by making use of its intermodal services and minimize the carbon dioxide, hydrocarbon, particle and nitrogen emissions. The total carbon dioxide emission reduction and diesel savings in the first quarter of 2016 are shown in Figure 6.

The data utilized in the case study have been given in Baykasoğlu and Subulan (2016). In this paper, these data are also elaborated in Tables 1–3 so as to present the uncertain parameters in a more detailed manner. In Table 1, fixed ownership costs, purchasing costs for new trucks/trailers and sale prices of second-hand trucks/trailers are given. The triangular fuzzy numbers which are related to the marine and railway transportation

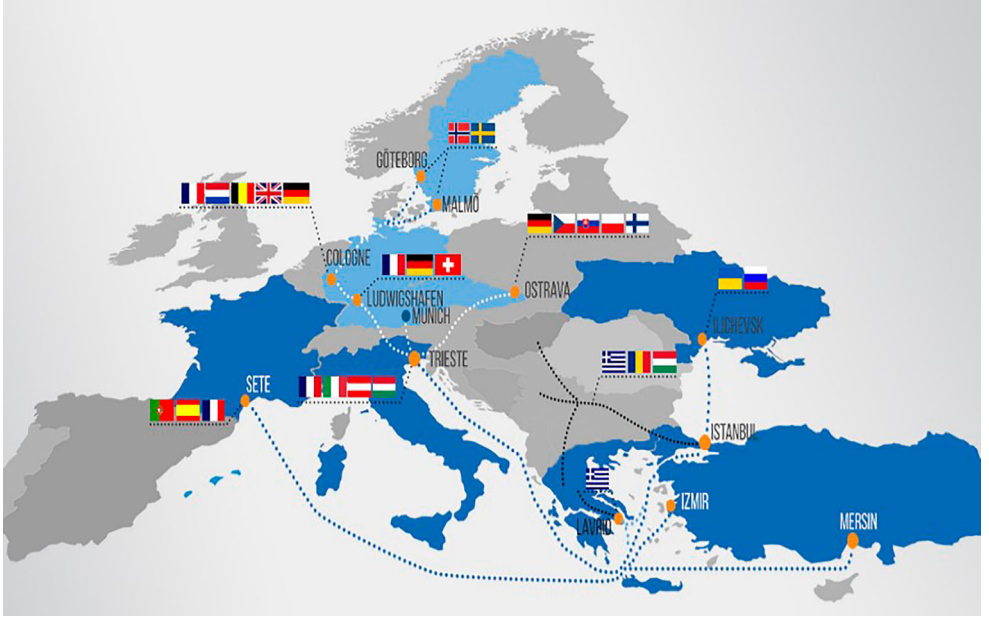


Figure 5. Intermodal logistics network representation of the real-world application (Baykasoğlu and Subulan 2016).

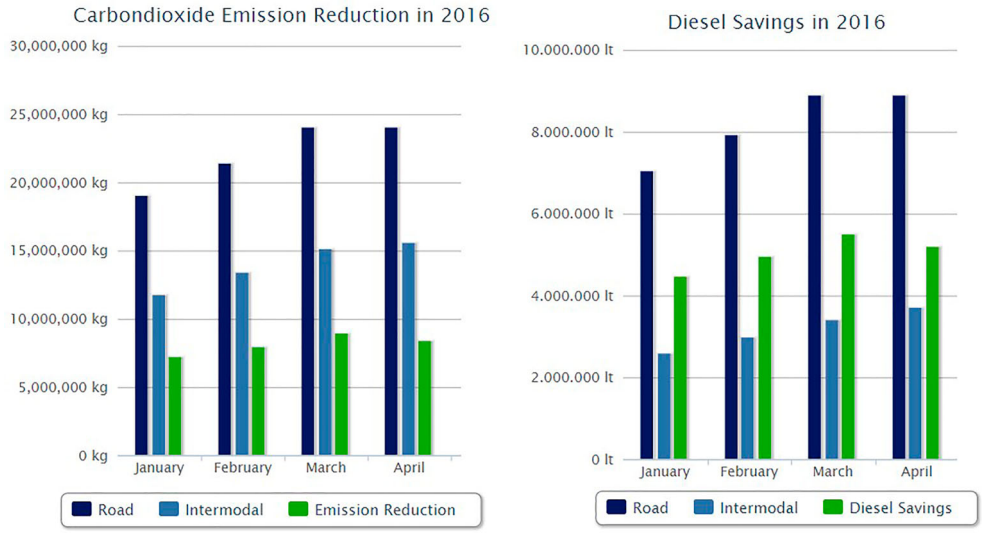


Figure 6. Environmental data: CO_2 emission reduction and diesel savings in 2016 (EKOL 2016).

Table 1. Fuzzy ownership, purchasing costs and sales prices of trucks and trailers.

	Trailers	European plated trucks	Turkish plated trucks
Ownership cost (€)	(200, 300, 500)	(1700, 2000, 2300)	(1500, 1700, 1900)
Purchasing cost (€)	(20,000, 30,000, 35,000)	(120,000, 150,000, 180,000)	(100,000, 120,000, 140,000)
Sales price (€)	(10,000, 15,000, 20,000)	(60,000, 75,000, 90,000)	(50,000, 60,000, 70,000)

Table 2. Fuzzy transport costs, capacities and transit times in railway and marine transportation.

By company-owned Ro-Ro vessels (€/trailer)	Transportation costs (€/trailer)	Transport capacity (trailer)	Marine and rail transit times (hrs.)
Istanbul – Lavrion	(450, 475, 500)	(200, 240, 250)	(15, 19, 22)
Istanbul – Trieste	(700, 750, 800)	(200, 240, 250)	(60, 63.75, 65)
Izmir – Sete	(1000, 1400, 1500)	(150, 165, 200)	(40, 48, 52)
By other logistics service providers' Ro-Ro vessels (€/trailer)			
Istanbul – Trieste	(750, 760, 800)	Unlimited	(60, 63.75, 65)
Istanbul – Toulon	(900, 918, 950)	Unlimited	(72, 88, 90)
Izmir – Trieste	(630, 648, 700)	Unlimited	(60, 72, 80)
Mersin Trieste	(750, 787, 820)	Unlimited	(90, 96, 100)
Railway trips by block trains (€/trip)			
Trieste – Ostrava	(16,000, 16,608, 17,000)	28	(20, 24, 26)
Trieste – Ludwigshafen	(20,000, 20,256, 21,000)	32	(17, 20, 22)
Trieste – Cologne	(23,000, 23,328, 24,000)	32	(20, 24, 26)
Railway trips by public trains (€/trailer)			
Trieste – Munich	(650, 685, 700)	Unlimited	(18, 22, 24)
Cologne – Malmö	(650, 685, 700)	Unlimited	(20, 22, 24)
Cologne – Gothenburg	(650, 685, 700)	Unlimited	(20, 22, 24)

costs, transport capacities and transit times are presented in Table 2. The marine/rail transit times between terminals are also given in Table 2. The transit times are also represented by fuzzy numbers because of the uncertainty in travel times, awaiting times at terminals and availabilities of Ro-Ro vessels, block and public trains. Other fuzzy data

Table 3. Other fuzzy data on environmental impact values, outsourcing option, availabilities, etc.

Uncertain parameters	Triangular fuzzy number
CO_2 emission value by road freight transportation (kg/km)	(0.8, 0.918, 1.0)
CO_2 emission value by marine transportation (kg/km)	(0.3, 0.357, 0.5)
CO_2 emission value by railway transportation (kg/km)	(0, 0.05, 0.1)
CO_2 emission value by empty truck repositioning (kg/km)	(0.8, 0.85, 0.9)
Ratio of road freight transportation for outsourcing option (%)	(0.7, 0.75, 0.8)
Ratio of combined marine and road transports for outsourcing option (%)	(0.2, 0.25, 0.3)
Number of monthly working days at the order countries (day/month)	(20, 25, 28)
Availability rate for trailers (%)	(60, 70, 90)
Availability rate for European plated trucks (%)	(60, 70, 90)
Availability rate for Turkish plated trucks (%)	(60, 70, 90)
Lower bounds on the number of block train services (trip/month)	(6, 8, 10)
Road freight transportation cost for a unit of loaded trailer (€/km)	(1.1, 1.2, 1.3)
Road freight transportation cost for a unit of empty trailer (€/km)	(1, 1.1, 1.2)
Road freight transportation cost for a unit of empty truck (€/km)	(0.8, 1.0, 1.1)

related to the environmental impact values, the composition of the outsourcing option, truck/trailer availabilities, number of working days at the order countries, loaded/empty road transportation costs, etc. are shown in Table 3.

4.2. Fuzzy multi-objective optimization results of the proposed intermodal fleet planning model

The computational results of the proposed fuzzy-stochastic model are presented separately for each objective function as a payoff matrix in Table 4. This payoff matrix consisting of the expected values of the fuzzy goals is used by the company experts to construct the membership functions while handling these conflicting objectives simultaneously. The computational experiments have shown that overall transport cost, transit time and CO_2 emission values will increase when random and fuzzy model parameters are taken into account in an integrated manner. Besides, when sample size (N), probability values for the chance constraints and the feasibility degree (α) of the fuzzy constraints are increased, all objective function values will increase due to the higher level of stochastic demand satisfaction and lower level of risk related to violations in the fuzzy constraints. The effects of stochastic freight demand on the objective values are more apparent than the fuzzy parameters. The more reliable and risk-free fleet plans can be provided in the case of a larger sample size, higher probability and feasibility degree. A weighted additive fuzzy goal programming approach is employed because of its simplicity and ease of use in order to solve the developed model with several conflicting objectives. Therefore, numerous compromise solutions can be offered to the transportation company under different uncertainty levels. The objective weights are assigned by the company experts as 0.425, 0.334 and 0.241 for total transport cost, transit time and environmental goals, respectively. In reality, these relative importance weights are determined based on desirable achievement degrees or overall priorities under a fuzzy group decision making environment (Baykasoğlu and Subulan 2016). In summary, the proposed fuzzy multi-objective model can be converted into a single objective crisp equivalent form as in Equation (158) via weighted additive

Table 4. Corresponding payoff matrix with expected values of fuzzy goals under a 24-hour runtime limit ($N = 5$, $P = 90\%$).

Feasibility degree, α	Objectives	Total transport cost (€)	Total transit time (h)	Total CO_2 emission (kg)	Elapsed Runtime (sec.)	Number of iterations	Optimality gap (%)
0	Total transport cost	**208,368,155.8**	10,736,858.7	141,836,614.5	20,188.68	3,224,705	0.0
	Total transit time	167,466,9842.01	**10,037,396.5**	137,788,578.2	2633.58	507,807	0.0
	Total CO_2 emission	164,075,9687.41	10,637,771.95	**127,504,456.2**	86,400	67,926,733	0.00,098
0.2	Total transport cost	**209,419,589.7**	10,737,214.4	141,614,404.03	17,342.22	3,393,198	0.0
	Total transit time	193,802,1117.6	**10,048,018.1**	138,228,368.6	86,400	24,693,484	0.00,099
	Total CO_2 emission	143,840,8811.8	10,656,699.5	**127,986,123.9**	86,400	2,600,101	0.001
0.4	Total transport cost	**210,419,679.4**	10,735,874.6	141,659,287.9	21,336.49	4,279,650	0.0
	Total transit time	169,754,8125.4	**10,054,329.5**	138,394,970.3	86,400	9,575,045	0.000,995
	Total CO_2 emission	165,162,0834.03	10,664,107.4	128,274,162.2	86,400	52,341,878	0.0021
0.6	Total transport cost	**211,822,490.08**	10,744,465.4	142,426,918.7	22,034.72	4,103,784	0.0
	Total transit time	187,583,5216.8	**10,062,419.4**	138,838,314.02	86,400	18,632,302	0.0065
	Total CO_2 emission	143,898,0115.3	10,677,337.08	**128,645,947.3**	86,400	6,537,921	0.0086
0.8	Total transport cost	**213,275,778.8**	10,768,660.5	143,511,196.8	25,092.01	4,363,353	0.0
	Total transit time	194,696,1293.1	**10,069,571.06**	139,102,413.8	86,400	15,404,999	0.0054
	Total CO_2 emission	167,604,3936.04	10,693,201.5	128,960,638.2	86,400	4,049,039	0.0177
1.0	Total transport cost	**215,180,409.6**	10,772,720.2	144,393,276.7	86,400	11,272,108	0.011
	Total transit time	187,242,2461.8	**10,084,612.1**	139,643,963.1	86,400	15,809,425	0.00,962
	Total CO_2 emission	145,270,1834.7	10,717,087.2	**129,527,566.3**	86,400	15,164,830	0.02

fuzzy goal programming. Indeed, the following weighted additive model is formulated for a specific feasibility degree ($\alpha = 0.4$):

$$\begin{cases} \text{Maximize} \quad 0.425 \times \mu_{Cost} + 0.334 \times \mu_{Transit-time} + 0.241 \times \mu_{CO2} \\ \mu_{Cost} \leq \dfrac{278559021.3 - Z_{Cost}}{68139341.9} \\ \mu_{Transit-time} \leq \dfrac{10054329.5 - Z_{Transit-time}}{681545.1} \\ \mu_{CO2} \leq \dfrac{141649278.9 - Z_{CO2}}{13385116.7} \\ \sum_{k=1}^{3} w_k = 0.425 + 0.334 + 0.241 = 1 \qquad 0 \leq w_k \leq 1 \\ \text{Probabilistic,} \quad \text{Fuzzy and Deterministic Constraint sets (33)} - (144) \\ 0 \leq \mu_{Cost},\ \mu_{Transit-time},\ \mu_{CO2} \leq 1 \end{cases} \tag{158}$$

After performing multi-objective optimization, compromise solutions can be computed and are shown in Tables 5 and 6 for different feasibility degrees of constraints. In Table 5, the objective values associated with these compromise solutions are provided. The associated degrees of satisfaction of the fuzzy goals are also given in Table 6. Among these solutions, a balanced solution between the feasibility degree of constraints and the global satisfaction degree should be found. Company experts are therefore asked to specify aspiration levels of the total cost, transit time, CO2 emission values and their tolerance thresholds. Then, the general membership functions are constructed independently of the feasibility degree by the experts.

In detail, it is assumed that the company experts will be fully satisfied with a cost objective value lower than €193,052,782.1 and they will be dissatisfied with a cost value which is more than €278,559,021.3. Similarly, they will be fully satisfied with a total transit time value which is lower than 10,037,396.5 h and dissatisfied with a transit time value of more than 10,772,720.2 h. Furthermore, it is assumed that they will be fully satisfied with the environmental impact value lower than 127,504,456.2 kg and dissatisfied with

Table 5. Fuzzy multi-objective optimization results under a 24-hour runtime limit ($N = 5$, $P = 90\%$).

Feasibility degree, α	Total transport cost (€)	Total transit time (hrs)	Total CO_2 emission (kg)	Optimality gap (%)
0.0	223,905,622	10,296,311.8	136,027,055.7	1.38
0.2	224,061,184.9	10,295,298.8	136,388,944.2	0.0655
0.4	224,879,894.8	10,293,739.5	136,822,122.1	0.0653
0.6	225,838,131.7	10,296,600.4	137,288,642.1	0.074
0.75	225,030,213.7	10,327,314.8	137,544,296.1	0.186
1.0	227,121,521.1	10,348,781.8	138,402,596	1.51

Table 6. Results via weighted additive fuzzy goal programming approach ($N = 5$, $P = 90\%$).

Feasibility degree, α	Global Weighted Satisfaction Degree ($w_1 \cdot \mu_{z1} + w_2 \cdot \mu_{z2} + w_3 \cdot \mu_{z3}$)	Satisfaction Degree μ_{z1}	Satisfaction Degree μ_{z2}	Satisfaction Degree μ_{z3}
0.0	0.6389770	0.7786398	0.6298367	0.4053513
0.2	0.6415665	0.7882309	0.6412044	0.3834277
0.4	0.6385760	0.7877846	0.6487247	0.3613837
0.6	0.6449223	0.7899855	0.6566493	0.3728530
0.75	0.6581661	0.8199471	0.6313150	0.4100805
1.0	0.6478208	0.8115908	0.6160927	0.4029865

Table 7. Calculation of the fuzzy decision for finding a balanced compromise solution ($N = 5$, $P = 90\%$).

Feasibility degree, α	Global Satisfaction Degree μ_{z1}	Global Satisfaction Degree μ_{z2}	Global Satisfaction Degree μ_{z3}	Fuzzy Decision
0.0	0.639	0.699	0.495	0.0
0.2	0.637	0.7	0.473	0.1206
0.4	0.627	0.702	0.448	0.2369
0.6	0.616	0.698	0.421	0.347
0.75	0.626	0.6535	0.4055	0.4212
1.0	**0.601**	**0.622**	**0.354**	**0.5256**

a total CO_2 emission value which is more than 144,393,276.7 kg. Calculation of the compatibility index or global satisfaction degree of each α-parametric acceptable solution with respect to the above aspirations of the company can be made and shown in Table 7. This shows that the best compromise choice would be the solution provided by solving the weighted additive model corresponding to $\alpha = 1.0$ level in which the greatest membership degree has occurred.

4.3. Numerical results, comparisons and managerial implications

In this section, numerical results of the previously acquired compromise solution where feasibility degree, scenario size and probability is set to 1.0%, 5% and 90% are given and compared to the outcomes of the deterministic, stochastic and fuzzy-stochastic optimization cases.

4.3.1. Multi-mode load planning or intermodal freight transportation planning

The usage rates of different transportation modes and outsourcing are given in Figure 7. It should be emphasized here that intermodal transportation is the most preferred option for import/export transportation services. According to Figure 8, the marine and road freight transport costs constitute the largest portion of overall annual transport costs. The monthly variation associated with the usage rate of the outsourcing option is also presented for both import and export transportation operations in Figure 9.

When components of transit time and environmental goals are examined, it is recognized that intermodal transportation has the largest transit times and can be seen as the

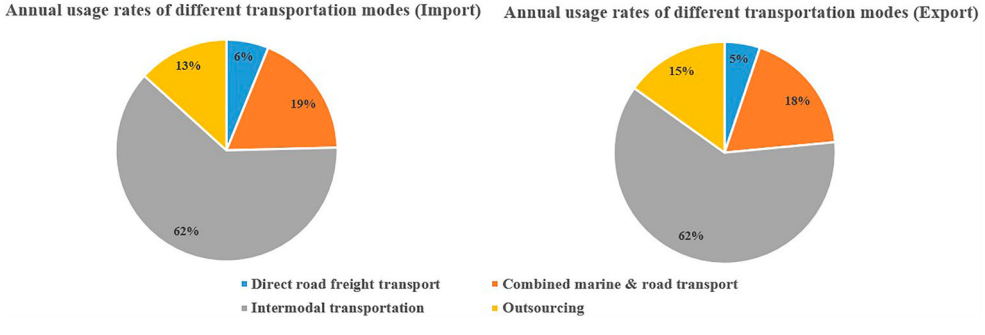


Figure 7. Usage rates of different transport modes in compromise solution ($\alpha = 1$, $N = 5$ and $P = 0.9$).

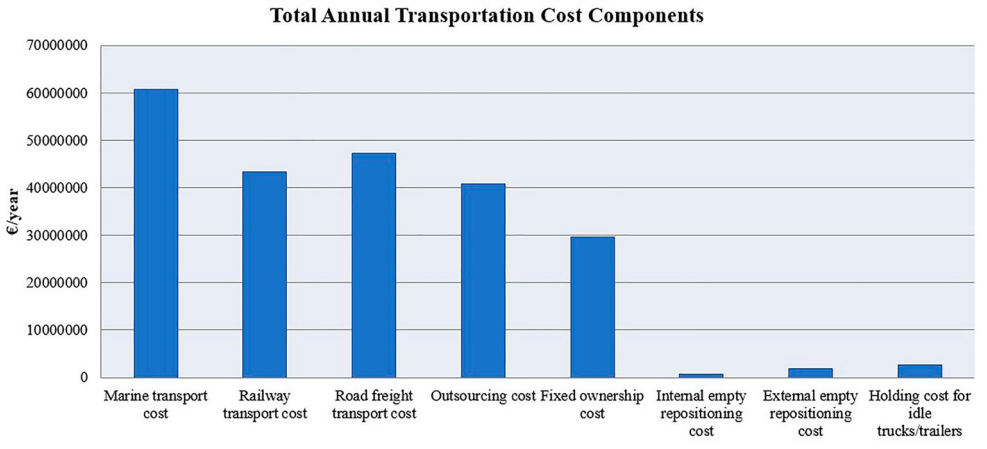


Figure 8. Comparison of the main components of total transport cost objective.

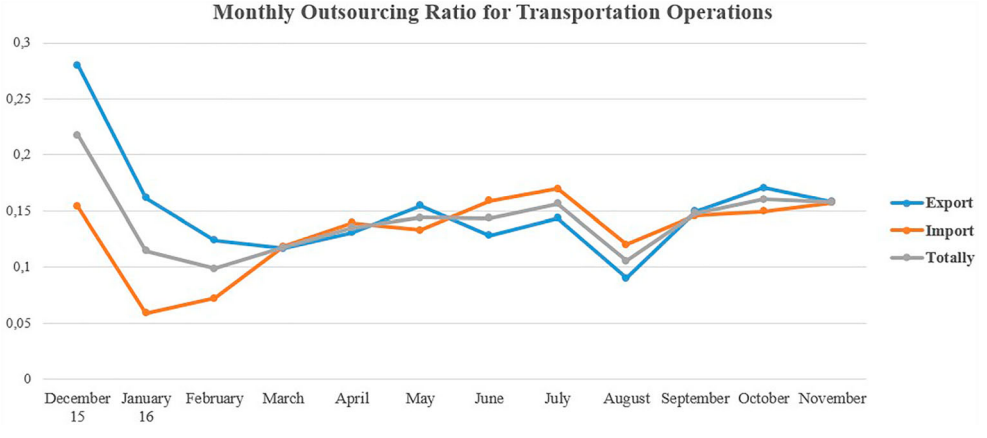


Figure 9. Monthly variation of the outsourcing ratio for import/export transportation operations.

lowest speed transportation mode. The combined marine and road freight transportation and outsourcing option are other transportation alternatives which may cause significant increases in total transit times.

For the total CO_2 emission objective, computational experiments have shown that rail transport is the most environmentally acceptable form of freight transportation. The effects of empty truck/trailer reposition on these objectives are also examined. It can be mentioned here that only minor increases may exist in total cost, transit time and environmental goals when empty flows are included. However, direct road freight transport, marine transport and outsourcing services may have substantial influences on total CO_2 emissions since road/marine transport have the largest carbon dioxide, hydrocarbon, particle and nitrogen emission values. Finally, load planning results in marine and railway transport are presented in Tables 8 and 9. According to Table 8, most freight should be transferred through the company-owned Ro-Ro vessels on the marine link between Istanbul and Trieste ports. Izmir-Trieste and Izmir-Sete are the other important marine links

Table 8. Load planning in maritime transportation.

Months	Dec. 15	Jan. 16	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.
The import/export load quantities in marine transport by the company-owned Ro-Ro vessels												
Istanbul/Lavrion	0	41	134	45	41	70	43	45	51	47	41	44
Istanbul/Trieste	1922	2420	2455	2530	2574	2420	2707	2530	2284	2530	2530	2530
Izmir/Sete	239	265	269	255	267	264	270	230	179	260	282	314
Lavrion/Istanbul	0	67	133	45	41	39	19	18	46	15	19	18
Trieste/Istanbul	2017	2329	2430	2530	2530	2530	2530	2498	2363	2579	2530	2530
Sete/Izmir	286	284	315	324	291	299	277	262	222	281	326	318
The import/export load quantities in marine transport by other logistics service providers' Ro-Ro vessels												
Istanbul/Trieste	0	53	0	148	0	195	0	21	0	75	202	195
Istanbul/Toulon	0	153	175	181	176	173	179	172	157	175	197	172
Izmir/Trieste	162	241	243	266	314	246	271	268	290	242	277	275
Mersin/Trieste	67	76	80	77	77	77	76	72	64	78	78	73
Trieste/Istanbul	0	0	0	198	38	85	133	0	0	0	130	52
Toulon/Istanbul	0	91	134	147	144	173	145	130	133	151	140	144
Trieste/Izmir	25	53	42	45	43	44	47	41	37	43	43	46
Trieste/Mersin	0	9	9	9	9	12	12	9	8	9	9	11

Table 9. Load planning in railway transportation.

Months	Dec. 15	Jan. 16	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.
Number of trips for block train services												
Trieste/Ostrava	12	10	11	11	11	11	11	11	11	12	11	11
Trieste/Ludwig.	23	23	24	23	25	23	25	24	24	25	23	25
Trieste/Cologne	36	33	37	36	36	36	37	37	36	36	36	37
Ostrava/ Trieste	12	10	11	11	11	11	10	11	11	12	11	11
Ludwig./ Trieste	24	22	24	24	24	24	25	24	24	25	23	25
Cologne/ Trieste	36	33	36	36	36	36	38	36	36	37	35	37
The import/export load quantities transferred by public trains												
Trieste/Munich	0	381	243	388	305	398	393	288	178	299	644	587
Cologne/Malmö	0	0	0	0	0	0	0	0	0	0	0	0
Cologne/Guthe.	63	71	118	96	111	100	89	128	156	129	163	176
Munich/ Trieste	4	264	217	370	273	372	360	252	182	262	591	434
Malmö/Cologne	0	0	0	0	0	0	0	0	0	0	0	0
Guthe./Cologne	62	88	85	90	101	102	100	93	89	88	88	95

where considerable amounts of freight are transferred. The number of chartered block trains and load quantities transported by public trains are presented in Table 9. It is clear that most freight is transferred by block trains from Trieste port to Cologne or Ludwigshafen stations. Furthermore, many loads are transported on the rail line between Trieste and Munich by public trains. It is also shown in Figure 10 that the usage rate of intermodal transportation may increase when stochasticity and fuzziness of some model parameters are taken into consideration.

Contrary to the single objective optimization, it is also revealed that the number of loads which are shipped by direct road freight transport services will grow in the case of fuzzy multi-objective optimization.

4.3.2. Truck/trailer fleet size optimization and fleet allocation

Firstly, loaded truck positions between the different locations, hubs and terminals are presented in this section. Figure 11 displays the number of loaded truck positions by European plated trucks between railway stations in Europe and order countries. Similarly, the number of loaded truck positions by Turkish plated trucks between seaports in Turkey and order countries are shown in Figure 12. Based on the computational

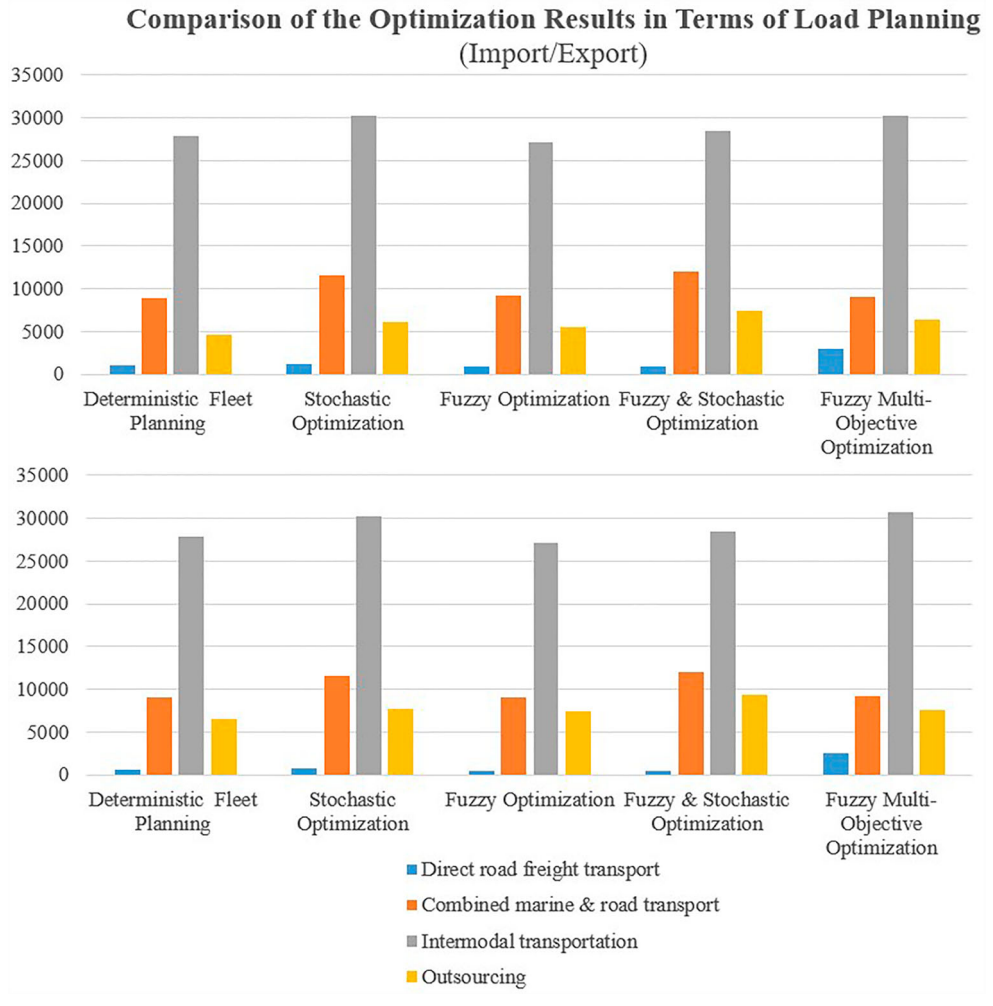


Figure 10. Comparison of single and multi-objective load planning results under uncertainty.

results, the majority of the loaded truck positions are performed by European plated trucks between seaports in Europe, railway terminals and order countries. In addition, direct road freight transportation operations are mostly conducted by Turkish plated trucks between seaports in Turkey and order countries.

The monthly truck and trailer fleet size requirements are also depicted in Figures 13 and 14. Most of the European plated trucks should operate between railway stations and the order countries. Furthermore, it is also shown that the total trailer fleet size is larger than the truck fleet size as an expected result. Finally, most trailers should be allocated to seaports in Turkey rather than order countries.

In Figure 15, fleet allocation among seaports in Europe and order countries is visualized. Most European plated trucks should be allocated to seaports and railway stations in Europe rather than order countries. When internal empty flows are evaluated, there is no empty trailer reposition between seaports in Turkey and order countries because of long travel distances. On the other hand, the volume of European plated empty truck flows

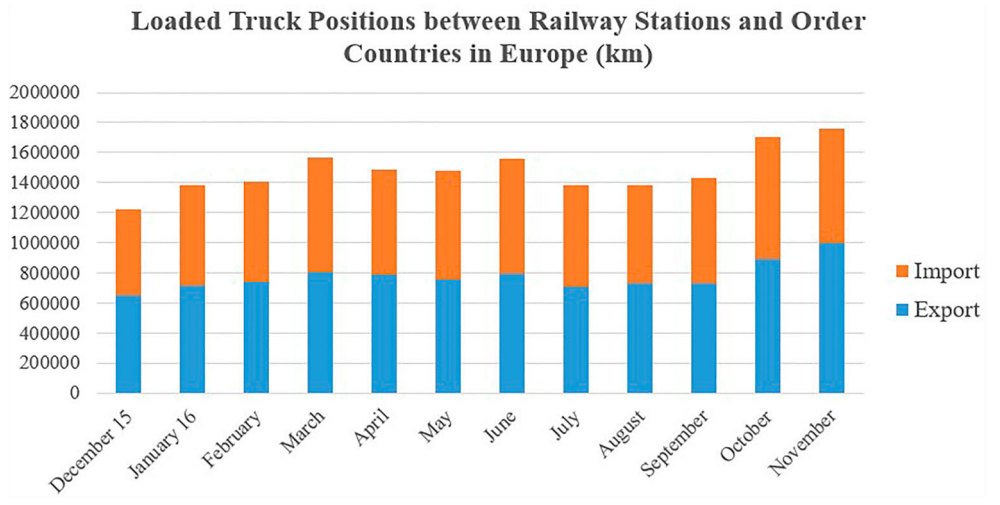


Figure 11. Loaded truck positions (in km) by EU plated trucks between railway stations and order countries.

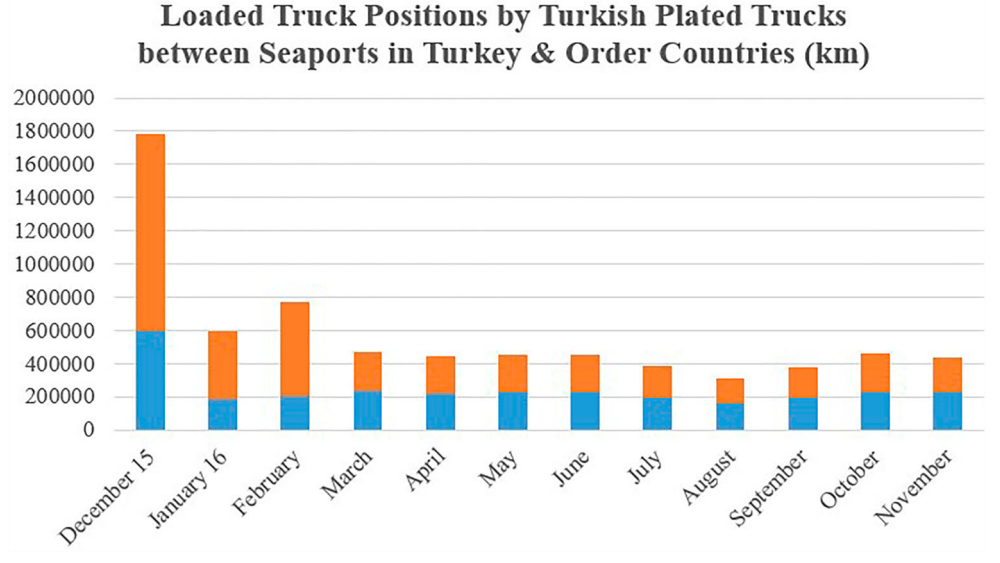


Figure 12. Loaded truck positions (in km) by Turkish plated trucks between seaports in Turkey and order countries.

among order countries and railway stations is relatively higher. The majority of European plated empty trucks are redistributed from order countries to rail stations as in Figure 16. When compared to the deterministic case, both truck and trailer fleet size requirements will increase under probabilistic freight demands. In addition, the fuzziness of some model parameters may cause increases in truck/trailer fleet sizes. Besides the transportation cost objective, both truck and trailer fleet sizes will increase radically

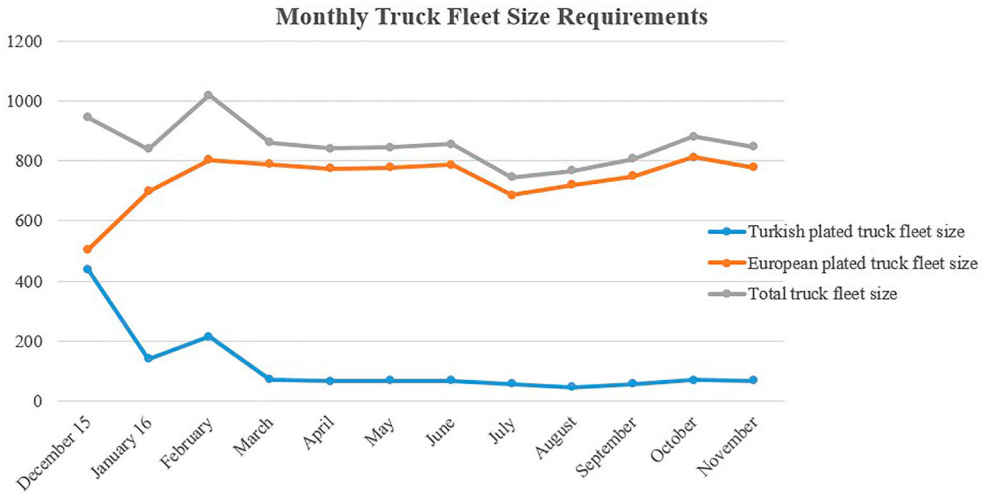


Figure 13. The monthly truck fleet size requirements with different plates on the road network.

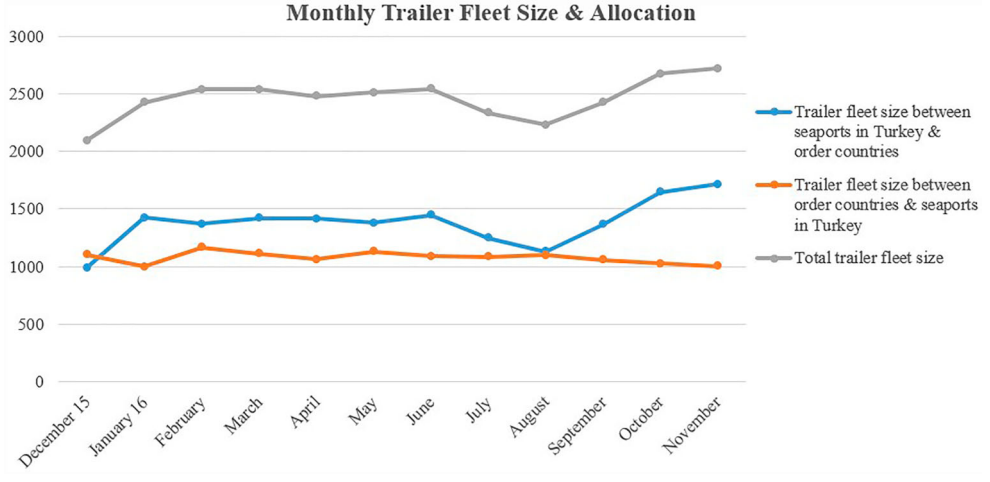


Figure 14. Monthly trailer fleet size requirements of different routes on the intermodal logistics network.

when the transit time and environmental goals are considered. The single objective and fuzzy multi-objective optimization results of trailer fleet sizing are compared in Figure 17.

4.3.3. Periodic truck/trailer inventories and fleet expansion/reduction decisions

The periodic truck and trailer inventories on different locations at the end of each month are shown in Figures 18 and 19. The truck inventory levels at the order countries and railway stations are comparatively higher than the seaports in Europe and Turkey. Most of the European plated truck inventories will accumulate at the order countries and seaports in Europe to the end of the planning horizon. In the compromise solution,

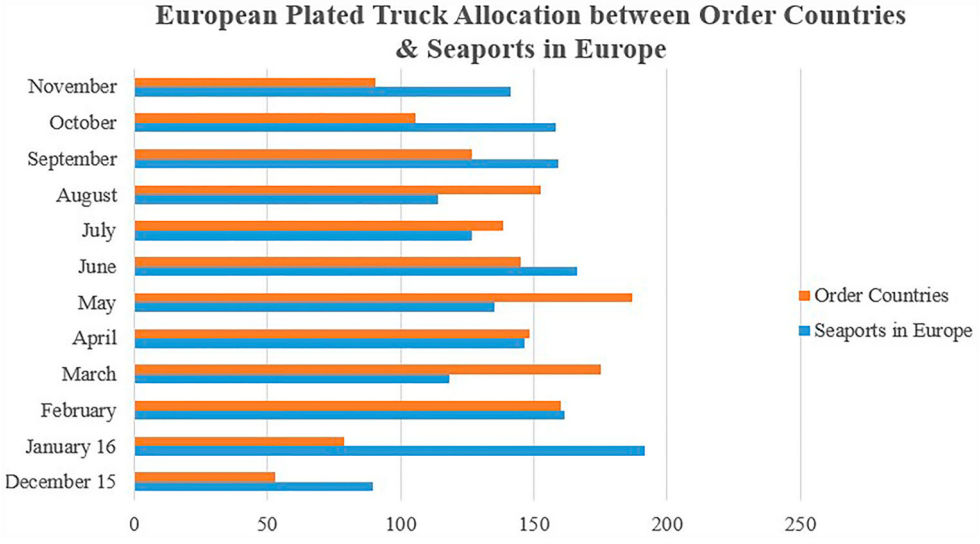


Figure 15. Monthly EU plated truck allocation between seaports in Europe and order countries.

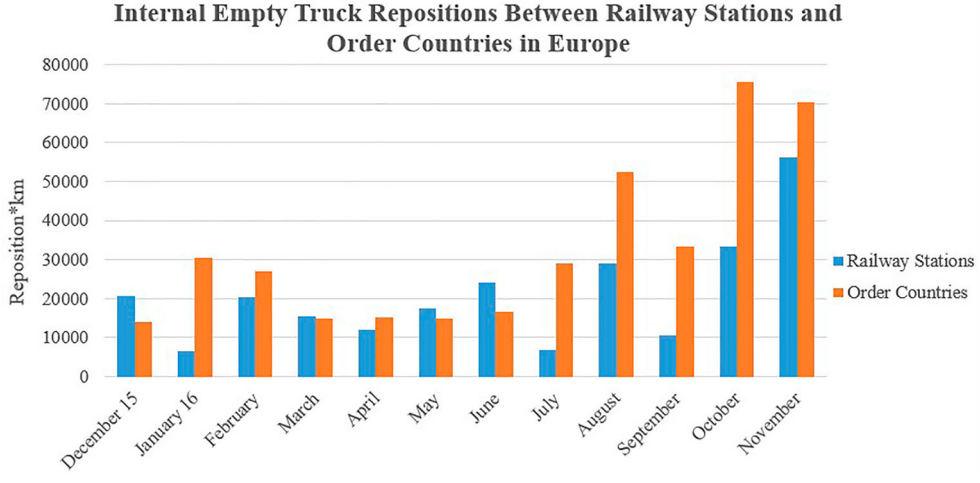


Figure 16. Monthly internal empty truck repositions (in km) between railway stations and order countries.

there are many idle or inactive trucks/trailers. In fact, these trucks/trailers waiting at different locations can be used to conduct empty flows within the monthly time period. The number of idle Turkish plated trucks for the first three months are quite high; on the other hand, there are no idle European plated trucks throughout the planning horizon. According to truck/trailer purchases and sales, most Turkish plated trucks could be sold at the beginning of the planning period. However, it should be noted here that the transportation company could purchase and sell European plated trucks at different locations in Germany so as to balance variations of fleet size requirements

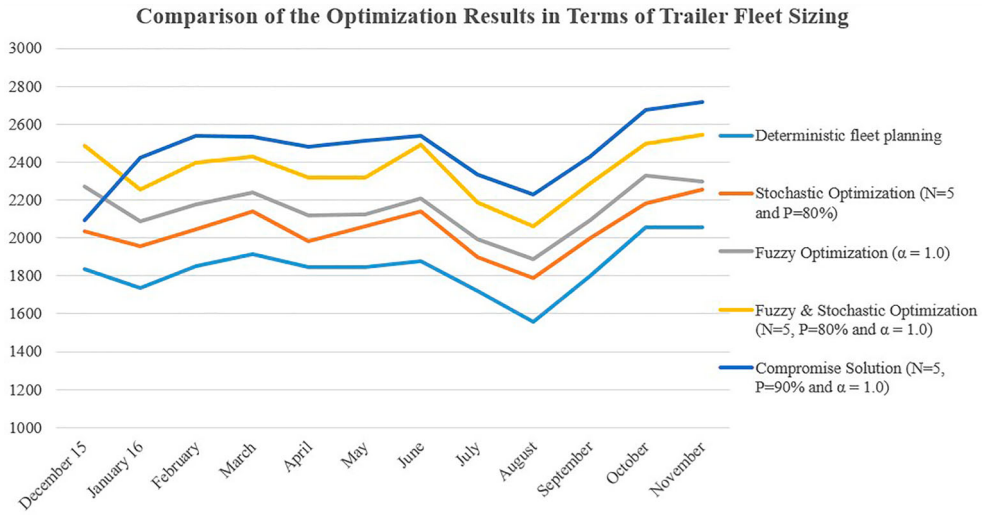


Figure 17. Comparison of single and multi-objective trailer fleet sizing results under uncertainty.

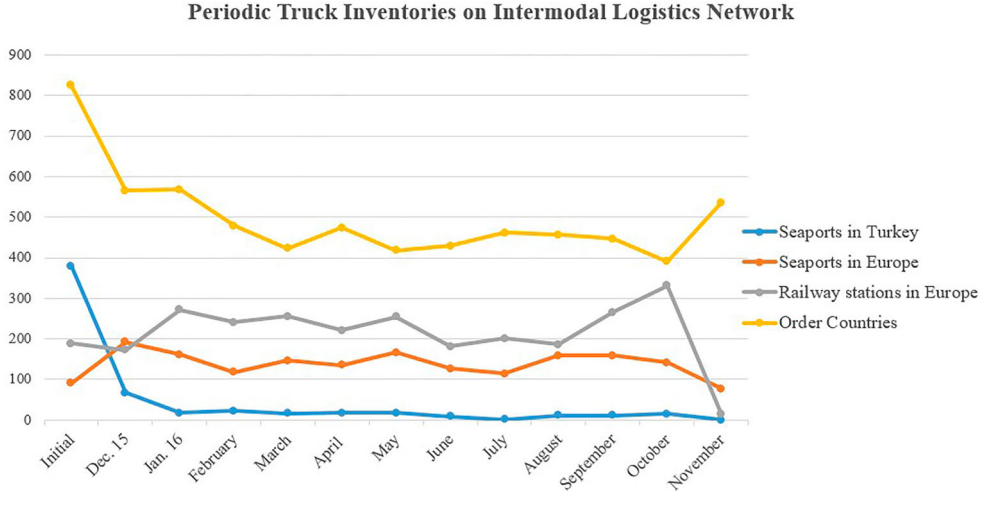


Figure 18. Periodic truck inventories at different nodes on the intermodal logistics network.

at different months. Additionally, the flexibility of truck/trailer purchases/sales may also reduce usage of the outsourcing option and empty truck/trailer repositions which are not considered desirable by the transportation company.

When all model parameters are assumed to be crisp, minimal inventory levels are yielded. In the case of single objective optimization, i.e. minimization of overall transport costs, the number of idle trucks and trailers may decrease when all uncertainties into the fleet planning problem are considered. As depicted in Figure 20, the compromise solution can provide minimal truck purchases/sales and trailer sales despite the maximal trailer purchasing amounts. In contrast to truck/trailer inventory levels, the number of truck purchases/sales may reduce when different types of uncertainties are considered.

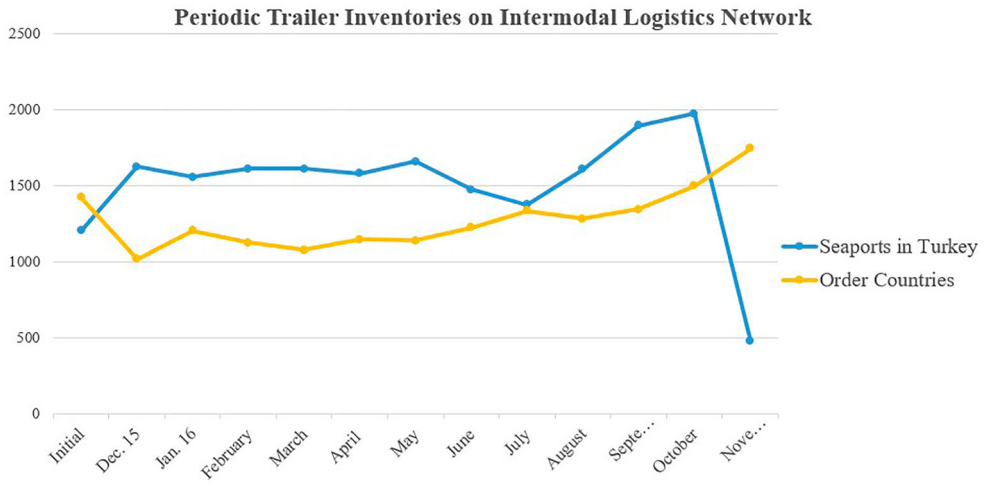


Figure 19. Periodic trailer inventories at seaports in Turkey and order countries in Europe.

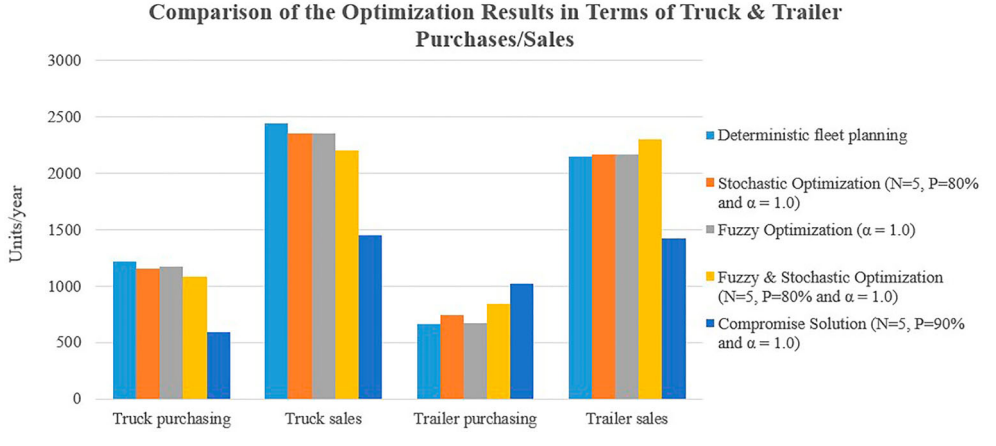


Figure 20. Comparison of single/multi-objective optimization results for truck/trailer purchases and sales.

4.3.4. External empty truck/trailer repositioning issues

The number of external empty truck/trailer repositions is portrayed in Figure 21. There are no external empty truck/trailer repositions among the seaports themselves in Turkey. European plated empty trucks are repositioned among seaports themselves, between seaports, railway stations and order countries. At the beginning of the year, several European plated empty trucks are redistributed from order countries to seaports in Turkey in order to perform loaded truck positions by direct road transportation. Figure 21 shows that a large part of the external empty trailer repositions could be realized with European plated trucks.

Both empty truck and trailer transfers are generally higher than the stochastic and fuzzy optimization cases (see Figure 22). Furthermore, it is expressed by company experts that

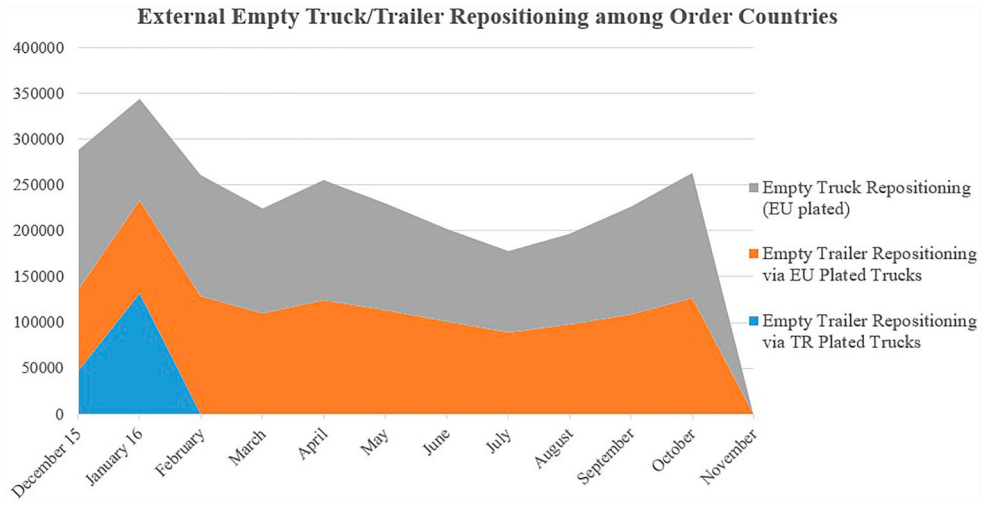


Figure 21. External empty truck and trailer repositions (in km) among order countries.

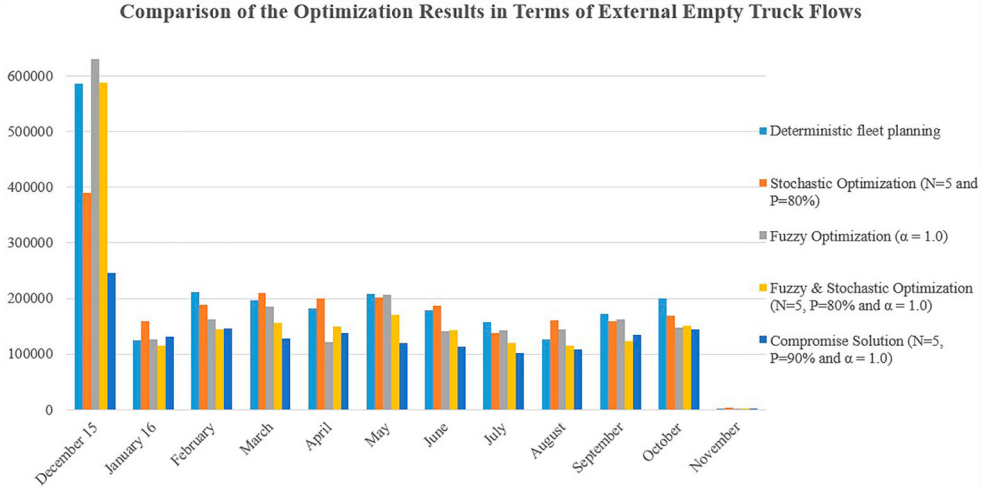


Figure 22. Comparison of single and multi-objective optimization results for external empty truck repositioning.

the compromise solution contains an acceptable level of external empty flows. Moreover, deterministic and fuzzy fleet planning models have the largest internal empty repositioning quantities. The compromise solution has the minimal level of internal empty truck flows which makes this solution more appropriate.

5. Conclusions

In this paper a new fuzzy-stochastic mixed-integer programming model has been proposed to solve a multi-objective, multi-mode, multi-resource and multi-period fleet

planning problem by considering simultaneously intermodal freight transportation planning, truck/trailer fleet sizing and composition, fleet expansion/reduction, truck/trailer inventories on the intermodal logistics network and empty repositioning issues. Different types of truck plates such as Turkish or European plated trucks and their associated cabotage limitations in Europe have also been incorporated within the proposed model while performing empty flows. Besides the economic objective, maximal customer service level satisfaction has also been targeted by minimizing total road, marine and railway transit times. Moreover, CO_2 emissions generated from the usage of different modes of transport have also been optimized so as to reduce the environmental impact. Different uncertainty categories such as randomness and fuzziness have been taken into consideration instantaneously via the hybrid fuzzy-stochastic approach. Finally, a weighted additive fuzzy goal programming approach was applied with the purpose of handling these conflicting objectives concurrently and obtaining compromise solutions under different uncertainty levels. The validity and practicality of the proposed model were presented based on a real-world application of a large-sized international transportation company in Turkey. The computational results have shown that external empty truck/trailer repositions can provide major reductions in the total transport cost goal although they do not have a substantial impact on the timing and environmental goals. It was also revealed that logistics companies can provide a sustainable competitive advantage by improving their fleet planning mechanisms. In the future, the holistic intermodal fleet planning model proposed here can be extended by integrating some operational level decisions such as truck routing, scheduling of block trains and Ro-Ro vessels.

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